The Double-Stranded Codon Torus: Duplex Mirror Fibration of DNA Coding Space, Markovian Lumpability, and Information-Theoretic Consequences

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Abstract

Biological DNA exists natively as an antiparallel double-stranded duplex, whereas standard mathematical models of coding sequences treat a gene as a single-stranded string and regard the complementary strand as passive. In this paper we construct a rigorous information-theoretic and Markovian theory of protein-coding DNA directly on the *double-stranded codon state space*. Once the four nucleobases are encoded by $\mathbb{Z}/4\mathbb{Z}$, the 64 triplets form the finite discrete torus $\mathbb{T}_4^3 = (\mathbb{Z}/4\mathbb{Z})^3$, and the physical codon duplex is the graph $\mathcal{D} = \{(c, \rho(c)) : c \in \mathbb{T}_4^3\}$ of the Watson–Crick reverse-complement operation $\rho(c) = -Rc + \mathbf{31} \pmod 4$. We prove that ρ is a fixed-point-free affine involution isometric with respect to both the mutational Hamming metric and the cyclic Lee metric, partitioning the 64 codons into 32 duplex mirror orbits and splitting the ambient space $\mathbb{R}^{64} = V_+ \oplus V_-$ orthogonally into a 32-dimensional mirror-even subspace V_+ (static Chargaff duplex parity, *form*) and a 32-dimensional mirror-odd subspace V_- (strand-asymmetric coding bias, *program*). While the single-strand genetic code surjection $\alpha : \mathbb{T}_4^3 \to \mathcal{A}$ (64 → 21) leaves a 43-dimensional synonymous simplex bundle ($61 - 20 = 41$ on sense codons), we prove that the **two-strand genetic code fibration** $\alpha_{\mathrm{dup}}(c) := (\alpha(c), \alpha(\rho(c)))$ maps the wobble position of the sense strand onto the leading position of the antisense strand, partitioning $\mathbb{T}_4^3$ into 48 singletons and exactly 8 binary doublets (56 joint sense–antisense channels) and collapsing the synonymous fiber dimension from 43 to $64 - 56 = 8$ (7 on sense codons). For any joint law of two consecutive codons, we prove: (i) total bigram mutual information is invariant under antiparallel strand inversion, $I(\rho(C_i); \rho(C_{i-1})) = I(C_{i-1}; C_i)$; (ii) exact tripartite and quadripartite chain-rule decompositions isolating protein dipeptide syntax $I(A_{i-1}; A_i)$, codon-to-next-amino-acid context $d_{2b} = I(C_{i-1}; A_i \mid A_{i-1})$ (which vanishes if and only if the amino-acid projection is Kemeny–Snell Markov lumpable), and the contextual synonymous program $S_{\mathrm{syn}} = I(C_{i-1}; C_i \mid A_i) \geq 0$ alongside its pairwise refinement $S_{\mathrm{syn}}^{(2)} = I(C_{i-1}; C_i \mid A_{i-1}, A_i) \geq 0$; and (iii) an exact **two-strand decomposition** $S_{\mathrm{syn}} = I(C_{i-1}; A_i^\rho \mid A_i) + I(C_{i-1}; C_i \mid A_i, A_i^\rho)$ showing that across nine reference genomes 74.1%–88.5% of the sense-strand synonymous program is carried by the antisense amino-acid channel $A_i^\rho = \alpha(\rho(C_i))$. We further prove that deterministic Max-CAI codon optimization ($C_i = \phi(A_i)$) annihilates all conditional non-dipeptide layers ($d_{2b} = S_{\mathrm{syn}} = S_{\mathrm{syn}}^{(2)} = 0$) while incurring a non-positive Kullback–Leibler cross-scoring penalty under a natural reference kernel, establish a chemical data-processing inequality $I(C_{i-1}; Q_i \mid A_{i-1}) \leq d_{2b}$, and prove non-negativity of both time-reversal entropy production $\sigma \geq 0$ and duplex strand-current divergence $\sigma_{\mathrm{dup}} \geq \frac{1}{2}\|J^{\mathrm{mir}}\|_1^2 \geq 0$. All theorems are corroborated analytically without Monte Carlo surrogates across nine RefSeq genomes (9,834,024 held-out transitions), the 100-kb human chromosome 17 locus `NC_000017.11:55154472-55254472`, and a comparative six-frame duplex forensics audit (`codon_torus_real_audit_v3_bnt162b2.py`, `v3.5`) spanning wild-type

SARS-CoV-2 Spike (`NC_045512.2`), the 3,825-nt Pfizer/BioNTech `BNT162b2` construct, the 3,828-nt Moderna `mRNA-1273` construct and its 3,819-nt NCBI vector record (`OR134578.1`), deterministic human Max-CAI/Max-GC3 designs, and constructive zero-RNG Viterbi first-order Markov compilers enforcing Corollary 4.4 duplex stop/start guards. All integer constants $(2, 3, 4, 6, 7, 8, 16, 19, 20, 21, 24, 32, 40, 41, 42, 43, 48, 54, 56, 60, 61, 62, 63, 64, 361, 400, 779, 860, 1681, 1849, 2460, 2709)$ are derived from exact set cardinalities, orbit counts, simplex dimensions, or contingency degrees of freedom; no numerological interpretation is used.

Keywords: discrete torus, genetic code fibration, codon usage bias, conditional mutual information, Markov lumpability, data-processing inequality, Max-CAI codon optimization, Markov entropy production, Chargaff mirror symmetry, contingency-table likelihood-ratio test.

1 Scope and anti-numerology policy

The statement that "coding DNA state space has the structure of a discrete torus" must be formulated with mathematical precision. We do not claim that the physical DNA double helix is a spatial torus in $\mathbb{R}^3$; the physical molecule is a helical polymer. The mathematical object studied in this paper is the *finite state space of triplets (codons)* in a fixed reading frame and the stochastic processes evolving on that state space.

Once the four nucleobases are identified with the cyclic group $\mathbb{Z}/4\mathbb{Z}$, the set of 64 codons becomes the finite abelian group

$$\mathbb{T}_4^3 := (\mathbb{Z}/4\mathbb{Z})^3 \cong \mathbb{Z}^3/4\mathbb{Z}^3,$$

which is the three-dimensional discrete torus of side length four (the Cartesian product of three 4-cycles). All theorems in this paper concern the algebraic, metric, spectral, fiber-bundle, and information-theoretic structure of $\mathbb{T}_4^3$.

To ensure complete mathematical rigor, the following policy is enforced throughout the paper:

(a) Every integer constant explicitly used in the paper (including 2, 3, 4, 6, 7, 8, 16, 19, 20, 21, 24, 32, 40, 41, 42, 43, 48, 54, 56, 60, 61, 62, 63, 64, 361, 400, 779, 860, 1681, 1849, 2460, and 2709) is derived explicitly from a set cardinality, a group order, a free group-orbit count, a reading-frame period, a contingency-table degree-of-freedom count, or the real dimension of a probability simplex. Measured empirical statistics (such as profile concordance R^2_{prof}) and genomic window coordinates are observational data, never postulated constants.

(b) No external arithmetic coincidence, cosmological constant, or particle-physics analogy is used as evidence.

(c) No physical or biological claim is made beyond the proved state-space identities, information-theoretic inequalities, and held-out statistical audits.

(d) Empirical verification is formulated analytically via held-out log-likelihoods and exact/asymptotic likelihood-ratio (G^2) contingency tests on real coding sequences, without relying on Monte Carlo surrogate simulations as mathematical proofs.

Remark 1.1 (What is proved and what is not proved)**.** We prove structural theorems about the double-stranded codon torus $\mathbb{T}_4^3$, closure semantics, the single-strand genetic-code fibration $\alpha : \mathcal{C} \to \mathcal{A}$ $(64 \to 21)$ and the two-strand duplex fibration $\alpha_{\text{dup}} : \mathcal{C} \to \mathcal{A} \times \mathcal{A}$ $(64 \to 56)$, the tripartite, quadripartite, and two-strand decompositions of codon-bigram mutual information, Markov lumpability, the collapse of conditional syntax under deterministic Max-CAI designs, the chemical data-processing hierarchy, and non-equilibrium Markov entropy production. In particular, the integers 43 and 8 have no metaphysical meaning: $43 = 64 - 21$ is the exact real dimension

$$\dim_{\mathbb{R}} \Delta_{\text{syn}} = \sum_{a \in \mathcal{A}} (k_a - 1) = 64 - 21 = 43$$

of the single-strand conditional synonymous probability bundle in the standard $64 \to 21$ genetic code, while $8 = 64 - 56$ is the exact real dimension of the two-strand duplex synonymous bundle Δ_{dup} induced by $\alpha_{\text{dup}}(c) = (\alpha(c), \alpha(\rho(c)))$ (Theorem 4.3).

2 The codon alphabet as a discrete torus

Let the nucleotide alphabet be

$$B = \{A, C, G, T\}.$$

Fix the reference encoding $e : B \to \mathbb{Z}/4\mathbb{Z}$ given by

$$e(A) = 0, \qquad e(C) = 1, \qquad e(G) = 2, \qquad e(T) = 3,$$

and identify B with $\mathbb{Z}/4\mathbb{Z}$. Under this identification, Watson–Crick complementary base pairing ($A \leftrightarrow T$, $C \leftrightarrow G$) is given by the affine reflection

$$\kappa(x) = 3 - x \pmod 4.$$

Indeed, $\kappa(0) = 3$, $\kappa(3) = 0$, $\kappa(1) = 2$, and $\kappa(2) = 1$.

Definition 2.1 (Codon torus)**.** The codon state space is $\mathcal{C} = B^3$. Under the coordinatewise encoding e, we identify $\mathcal{C}$ with the finite abelian group

$$\mathbb{T}_4^3 := (\mathbb{Z}/4\mathbb{Z})^3.$$

On $\mathbb{Z}/4\mathbb{Z}$, the point-substitution distance is the Hamming metric $d_H(x, y) := \mathbf{1}_{\{x \neq y\}} \in \{0, 1\}$, while the cyclic graph distance (measuring steps around the 4-cycle 0–1–2–3–0) is the Lee metric

$$d_{\mathrm{Lee}}(x, y) := \min\big(|x - y|,\, 4 - |x - y|\big) \in \{0, 1, 2\}.$$

These extend additively to $\mathbb{T}_4^3$:

$$d_H(c, c') = \sum_{j=1}^{3} \mathbf{1}_{\{c_j \neq c'_j\}}, \qquad d_{\mathrm{Lee}}(c, c') = \sum_{j=1}^{3} d_{\mathrm{Lee}}(c_j, c'_j).$$

Proposition 2.2 (Discrete torus structure)**.** *The group $\mathbb{T}_4^3 \cong \mathbb{Z}^3/4\mathbb{Z}^3$ is a three-dimensional discrete torus. Its Cayley graph with respect to the standard generators $\pm e_1, \pm e_2, \pm e_3$ is the periodic $4 \times 4 \times 4$ cubic lattice $C_4 \square C_4 \square C_4$ with shortest-path distance d_{Lee}, while its complete point-mutation graph (allowing any single-base substitution at each coordinate) is the Hamming graph $H(3, 4) = K_4 \square K_4 \square K_4$ with shortest-path distance d_H.*

Proof. The quotient $\mathbb{Z}^3/4\mathbb{Z}^3$ identifies two integer triples $u, v \in \mathbb{Z}^3$ if and only if $u_j \equiv v_j \pmod 4$ for $j = 1, 2, 3$. Each coordinate therefore forms a cycle graph C_4 of length 4 under $\pm 1 \pmod 4$. The Cartesian product $C_4 \square C_4 \square C_4$ is the periodic $4 \times 4 \times 4$ lattice on $4^3 = 64$ vertices, whose graph distance is the sum of coordinatewise cycle distances d_{Lee}. Replacing each coordinate cycle C_4 by the complete graph K_4 (where every distinct base pair is at distance 1) yields the Cartesian product $K_4 \square K_4 \square K_4 = H(3, 4)$, whose graph distance is the Hamming distance d_H. □

Corollary 2.3 (Framed coding sequences as high-dimensional torus points)**.** *For a fixed reading frame, a coding sequence of length n codons is an element of*

$$(\mathbb{T}_4^3)^n \cong (\mathbb{Z}/4\mathbb{Z})^{3n}.$$

Thus the state space of framed coding sequences of length n codons is a finite discrete torus of dimension $3n$ over $\mathbb{Z}/4\mathbb{Z}$ with cardinality $4^{3n} = 64^n$.

Proof. Immediate from the associativity of direct products of finite abelian groups: $((\mathbb{Z}/4\mathbb{Z})^3)^n \cong (\mathbb{Z}/4\mathbb{Z})^{3n}$. □

Remark 2.4 (Auxiliary status of the Lee metric)**.** The Hamming metric d_H is the intrinsic single-nucleotide-substitution metric independent of base ordering. The Lee metric d_{Lee} is an auxiliary cyclic metric induced by the chosen encoding $B \cong \mathbb{Z}/4\mathbb{Z}$. None of the main theorems of this paper—the fixed-point-free orbit count, the genetic-code fiber dimensions, the tripartite and quadripartite information decompositions, the Markov lumpability criterion, the Max-CAI collapse theorem, the chemical data-processing inequality, or the Markov entropy production theorems—depends on this auxiliary cyclic metric.

2.1 Two-stranded (duplex) basis of the theory

The state space $\mathcal{C} = B^3$ defined above describes *one* nucleotide strand. The biological carrier of genetic information, however, is the double-stranded duplex, whose two antiparallel strands are related exactly by the reverse-complement involution ρ of Section 2.2. We therefore take the duplex

$$\mathcal{D} := \{(c, \rho(c)) : c \in \mathcal{C}\},$$

i.e. the 32 mirror orbits of Corollary 2.6, as the primitive object of the theory, and regard a single strand as the image of $\mathcal{D}$ under the projection that forgets the antisense partner. This projection is precisely what discards the mirror-odd subspace V_-: a single-strand analysis can see the static compositional *form* (the V_+ / Chargaff second-parity and 32-mirror-pair component established by Chargaff, Prabhu, and Perez [3, 4, 5, 14, 15, 16, 17, 23]) but is structurally blind to the strand-asymmetric sequential *program* carried by V_-. Hence the decomposition $\mathbb{R}^{64} = V_+ \oplus V_-$ is not a technical device but the algebraic expression of two-strandedness: V_+ encodes the symmetry *between* the two strands, V_- encodes their *difference*, and every conditional information layer S_{syn}, $S^{(2)}_{\text{syn}}$, d_{2b} of the subsequent sections is measured on the duplex through ρ (the dynamic mirror current $\|J^{\text{mir}}\|_1$ of Proposition 8.4 is the directed two-strand flux asymmetry). No result in this paper assumes a single-stranded ontology; the torus $\mathbb{T}^3_4$ is the per-strand coordinate space, and the theory lives on the ρ-equivariant structure of the duplex.

2.2 Reverse complement as a fixed-point-free isometric involution

For a codon $c = (c_1, c_2, c_3) \in \mathbb{T}^3_4$, the biological antiparallel reverse-complement operation reads the complementary strand from 5′ to 3′, reversing the coordinate order and complementing each base:

$$\rho(c_1, c_2, c_3) := \big(\kappa(c_3), \kappa(c_2), \kappa(c_1)\big).$$

In the group $(\mathbb{Z}/4\mathbb{Z})^3$, this is the affine transformation

$$\rho(c) = -Rc + \mathbf{31} \pmod 4,$$

where $R(c_1, c_2, c_3) = (c_3, c_2, c_1)$ is the coordinate-reversal matrix and $\mathbf{1} = (1, 1, 1)$.

Theorem 2.5 (Algebraic, metric, and encoding-invariant properties of ρ)**.** *The reverse-complement map $\rho : \mathbb{T}^3_4 \to \mathbb{T}^3_4$ satisfies the following properties:*

(i) ***Involution:*** *$\rho^2 = \mathrm{id}_{\mathbb{T}^3_4}$.*

(ii) ***Fixed-point-free:*** *$\rho(c) \neq c$ for all $c \in \mathbb{T}^3_4$.*

(iii) ***Encoding invariance:*** *The fixed-point-free property holds for any odd oligomer length $2m+1$ ($m \geq 0$) and is a set-theoretic property independent of the chosen bijection $e : B \to \mathbb{Z}/4\mathbb{Z}$. Among the $4! = 24$ bijections $e : B \to \mathbb{Z}/4\mathbb{Z}$, exactly 8 make Watson–Crick complementation take the affine form $\kappa(x) \equiv 3 - x \pmod 4$.*

(iv) ***Dual metric isometry:*** *ρ is an isometry of $\mathbb{T}^3_4$ with respect to both the mutational Hamming metric and the cyclic Lee metric:*

$$d_H\big(\rho(c), \rho(c')\big) = d_H(c, c'), \qquad d_{\text{Lee}}\big(\rho(c), \rho(c')\big) = d_{\text{Lee}}(c, c') \qquad \forall\, c, c' \in \mathbb{T}^3_4.$$

Proof. (i) For any $c = (c_1, c_2, c_3) \in \mathbb{T}^3_4$,

$$\rho^2(c) = \rho(3 - c_3, 3 - c_2, 3 - c_1) = \big(3 - (3 - c_1),\, 3 - (3 - c_2),\, 3 - (3 - c_3)\big) = (c_1, c_2, c_3) \pmod 4.$$

(ii) Suppose $\rho(c) = c$. Then the middle coordinate c_2 must satisfy

$$c_2 \equiv 3 - c_2 \pmod 4 \iff 2c_2 \equiv 3 \pmod 4.$$

In $\mathbb{Z}/4\mathbb{Z}$, $2c_2 \in \{0, 2\}$, which is even and can never equal 3. Hence ρ has no fixed points.

(iii) Set-theoretically, prior to any numerical encoding, Watson–Crick complementation $\kappa : B \to B$ is a fixed-point-free involution on the 4-element alphabet B ($\kappa(b) \neq b$ for all $b \in B$). On B^{2m+1}, reverse complementation maps $(b_1, \ldots, b_{2m+1})$ to $(\kappa(b_{2m+1}), \ldots, \kappa(b_1))$. Any fixed point must satisfy $b_{m+1} = \kappa(b_{m+1})$ at the central coordinate $m+1$, which is impossible because κ has no fixed points on B. To represent κ as $x \mapsto 3 - x \pmod 4$, the unordered complementary pairs $\{A, T\}$ and $\{C, G\}$ must be assigned to the residue pairs $\{0, 3\}$ and $\{1, 2\}$ in $\mathbb{Z}/4\mathbb{Z}$. There are 2 ways to match the pairs, 2 ways to orient $\{0, 3\}$, and 2 ways to orient $\{1, 2\}$, giving $2 \times 2 \times 2 = 8$ valid encodings.

(iv) For any $x, y \in \mathbb{Z}/4\mathbb{Z}$, $|\kappa(x) - \kappa(y)| = |(3 - x) - (3 - y)| = |x - y|$, so κ preserves both d_H and d_{Lee} on $\mathbb{Z}/4\mathbb{Z}$. Since the coordinate reversal R permutes the three summands of $d_H(c, c')$ and $d_{\text{Lee}}(c, c')$, their sums are invariant under ρ. □

Corollary 2.6 (Partition into 32 mirror pairs and orthogonal parity decomposition [4, 5, 16, 17]). *(i) The group $\{\mathrm{id}, \rho\} \cong \mathbb{Z}/2\mathbb{Z}$ acts freely on $\mathbb{T}_4^3$, partitioning the 64 codons into exactly*

$$\frac{|\mathbb{T}_4^3|}{2} = \frac{64}{2} = 32$$

disjoint 2-element mirror orbits $M_p = \{c, \rho(c)\}$ ($p = 1, \ldots, 32$).

(ii) The induced linear operator $(\rho^* f)(c) := f(\rho(c))$ on the ambient vector space $\mathbb{R}^{\mathcal{C}} \cong \mathbb{R}^{64}$ is an orthogonal involution ($(\rho^*)^2 = I$, $(\rho^*)^{\mathsf{T}} = \rho^*$) that splits $\mathbb{R}^{64}$ into the orthogonal direct sum

$$\mathbb{R}^{64} = V_+ \oplus V_-, \qquad \dim_{\mathbb{R}} V_+ = \dim_{\mathbb{R}} V_- = 32,$$

where $V_+ = \{f : f(\rho(c)) = f(c)\}$ is the mirror-even subspace (static strand-parity form) and $V_- = \{f : f(\rho(c)) = -f(c)\}$ is the mirror-odd subspace (strand-asymmetric coding bias). For any probability distribution p on $\mathcal{C}$, the L_1 mirror asymmetry

$$\mathrm{Mirror}_{L_1}(p) := \frac{1}{2} \sum_{c \in \mathcal{C}} \big| p(c) - p(\rho(c)) \big| \in [0, 1]$$

equals the L_1-norm $\|P_{V_-} p\|_1$ of the projection $P_{V_-} p = \frac{1}{2}(p - \rho^* p)$, and vanishes if and only if $p \in V_+$.

Proof. Part (i) follows immediately from Theorem 2.5(i)–(ii). For part (ii), on each 2-element orbit $M_p = \{c, \rho(c)\}$, ρ^* acts as the 2×2 swap matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, which has eigenvalues $+1$ (eigenvector $(1, 1)^{\mathsf{T}}$) and -1 (eigenvector $(1, -1)^{\mathsf{T}}$). Summing over the 32 disjoint orbits yields $\dim_{\mathbb{R}} V_+ = \dim_{\mathbb{R}} V_- = 32$, with orthogonal projectors $P_{V_\pm} = \frac{1}{2}(I \pm \rho^*)$. Finally, $\|P_{V_-} p\|_1 = \sum_{c \in \mathcal{C}} \big| \frac{1}{2}(p(c) - p(\rho(c))) \big| = \mathrm{Mirror}_{L_1}(p)$. □

2.3 Spectral character theory and mirror block-diagonalization on $\mathbb{T}_4^3$

Because $G = \mathbb{T}_4^3 = (\mathbb{Z}/4\mathbb{Z})^3$ is a finite abelian group, its Pontryagin dual group $\widehat{G}$ consists of 64 characters $\chi_k : G \to \mathbb{C}^\times$ indexed by wavevectors $k = (k_1, k_2, k_3) \in (\mathbb{Z}/4\mathbb{Z})^3$:

$$\chi_k(c) := \exp\left(\frac{2\pi i}{4}\, k \cdot c\right) = i^{k_1 c_1 + k_2 c_2 + k_3 c_3}, \qquad \langle \chi_k, \chi_{k'} \rangle = \frac{1}{64} \sum_{c \in \mathbb{T}_4^3} \overline{\chi_k(c)} \chi_{k'}(c) = \delta_{k,k'}.$$

Proposition 2.7 (Spectral diagonalization and mirror block reduction). *Let $(T_P f)(x) := \sum_{y \in \mathbb{T}_4^3} P_{xy} f(y)$ be the Markov transition operator on $\mathbb{C}^{\mathbb{T}_4^3} \cong \mathbb{C}^{64}$.*

(i) If P is translation-invariant on the torus ($P_{xy} = K(y - x)$ for a kernel $K : \mathbb{T}_4^3 \to \mathbb{R}_{\geq 0}$), then the 64 characters χ_k are a complete orthonormal eigenbasis of T_P with eigenvalues given by the discrete Fourier transform $\widehat{K}(k) = \sum_{z \in \mathbb{T}_4^3} K(z) i^{k \cdot z}$.

(ii) For a general (inhomogeneous) Markov kernel P, T_P commutes with the mirror operator ρ^* ($\rho^* T_P = T_P \rho^*$) if and only if P is mirror-equivariant: $P_{\rho(x),\rho(y)} = P_{xy}$ for all $x, y \in \mathbb{T}_4^3$. Under this condition, V_+ and V_- are T_P-invariant subspaces, and T_P reduces to the block-diagonal form $T_P = T_+ \oplus T_-$ of two 32×32 operators.

Proof. (i) For $P_{xy} = K(y - x)$, $(T_P \chi_k)(x) = \sum_y K(y - x)\chi_k(y) = \sum_z K(z)\chi_k(x + z) = \widehat{K}(k)\chi_k(x)$. (ii) For any function f, $(\rho^* T_P f)(x) = \sum_y P_{\rho(x),y} f(y) = \sum_z P_{\rho(x),\rho(z)}(\rho^* f)(z)$, whereas $(T_P \rho^* f)(x) = \sum_z P_{x,z}(\rho^* f)(z)$. These coincide for all f if and only if $P_{\rho(x),\rho(z)} = P_{x,z}$. Commutation with the projector $P_{V_\pm} = \frac{1}{2}(I \pm \rho^*)$ then implies $T_P(V_\pm) \subseteq V_\pm$. □

3 Closure semantics: boundary-free state space versus closed translation cycles

The discrete torus structure of the codon alphabet must be carefully distinguished from periodic boundary conditions on an individual gene or transcript. The former is a property of the alphabet state space; the latter is a property of a trajectory in that space.

Definition 3.1 (Coordinate closure of the codon torus)**.** For any $c = (c_1, c_2, c_3) \in \mathbb{T}_4^3$ and the standard basis vectors e_1, e_2, e_3 of $(\mathbb{Z}/4\mathbb{Z})^3$,

$$c + 4e_j = c \pmod 4, \qquad j = 1, 2, 3.$$

Equivalently, opposite faces of the combinatorial cube $\{0, 1, 2, 3\}^3$ are identified. This is the exact sense in which the codon state space has no boundary.

Remark 3.2 (No state-space boundary does not imply closed trajectories)**.** A boundary-free state space admits open (non-closed) trajectories. A coding sequence of length n codons is a discrete trajectory

$$\gamma : \{0, 1, \ldots, n - 1\} \to \mathcal{C}.$$

Such a trajectory is a closed loop only if an additional periodic boundary condition $\gamma(n) = \gamma(0)$ is imposed. While periodic boundary conditions are natural for circular DNA molecules or circularized synthetic RNAs, a canonical linear mRNA coding sequence (CDS) is an open path bounded by translation initiation and translation termination.

3.1 Start and stop codons are not identified by the torus

In the standard nuclear genetic code, translation initiation is canonically specified by ATG, while translation termination is specified by TAA, TAG, and TGA. Under our reference encoding $A = 0, C = 1, G = 2, T = 3$:

$$\mathrm{ATG} = (0, 3, 2), \qquad \mathrm{TAA} = (3, 0, 0), \qquad \mathrm{TAG} = (3, 0, 2), \qquad \mathrm{TGA} = (2, 3, 0).$$

Clearly,

$$\mathrm{ATG} \notin \{\mathrm{TAA}, \mathrm{TAG}, \mathrm{TGA}\}.$$

Moreover, applying the reverse-complement involution ρ yields

$$\rho(\mathrm{ATG}) = \mathrm{CAT}, \qquad \rho(\mathrm{TAA}) = \mathrm{TTA}, \qquad \rho(\mathrm{TAG}) = \mathrm{CTA}, \qquad \rho(\mathrm{TGA}) = \mathrm{TCA},$$

none of which belongs to the termination set $\{\mathrm{TAA}, \mathrm{TAG}, \mathrm{TGA}\}$.

Proposition 3.3 (No canonical start–stop identification)**.** *Neither the torus coordinate closure $c + 4e_j = c$, nor the reverse-complement involution ρ, nor the genetic-code fibration $\alpha : \mathcal{C} \to \mathcal{A}$ induces an identification between initiation and termination codons in the standard nuclear genetic code.*

Proof. The coordinate closure $c + 4e_j = c$ identifies residues modulo 4 at each coordinate position within the same triplet; it does not identify distinct triplets in $\mathbb{T}_4^3$. Set-theoretically, ATG and $\{\mathrm{TAA}, \mathrm{TAG}, \mathrm{TGA}\}$ are disjoint subsets of B^3 under any bijection $e : B \to \mathbb{Z}/4\mathbb{Z}$. Under ρ, $\rho(\mathrm{ATG}) = \mathrm{CAT} \notin \alpha^{-1}(\mathrm{Stop})$ and $\mathrm{ATG} \notin \rho(\alpha^{-1}(\mathrm{Stop})) = \{\mathrm{TTA}, \mathrm{CTA}, \mathrm{TCA}\}$. Under α, $\alpha(\mathrm{ATG}) = \mathrm{Met} \neq \mathrm{Stop}$ in $\mathcal{A}$. Thus no canonical identification exists. □

3.2 What is universally closed: the reading-frame phase modulo 3

Although the start and stop triplets are distinct points of $\mathbb{T}_4^3$, every canonical coding sequence possesses an exact positional phase closure.

Proposition 3.4 (Reading-frame phase closure modulo 3)**.** *Let a spliced coding sequence consist of an integer number n of codons ($3n$ nucleotides). If the first nucleotide of the start codon has 1-based (or 0-based) coordinate index s and the first nucleotide of the terminal stop codon has coordinate index t, then*

$$t = s + 3(n-1), \qquad \textit{and therefore} \qquad t \equiv s \pmod 3.$$

Proof. Each of the $n-1$ codons preceding the terminal stop codon spans exactly 3 nucleotides, so $t - s = 3(n-1)$, which is divisible by 3. □

Thus, the universal mathematical closure between the beginning and end of a gene is the conservation of the reading-frame phase in $\mathbb{Z}/3\mathbb{Z}$, while the triplet coordinates themselves live in $(\mathbb{Z}/4\mathbb{Z})^3$.

3.3 Closed codon rings and the extended translation-recycling automaton

When modeling genuinely closed genomic or translational cycles, two distinct mathematical constructions are available:

1. **Closed codon rings ($\gamma : \mathbb{Z}/n\mathbb{Z} \to \mathcal{C}$):** For circular genomes, plasmids, or circular RNAs, a framed sequence of n triplets is a map $\gamma : \mathbb{Z}/n\mathbb{Z} \to \mathbb{T}_4^3$, which satisfies $\gamma(n) = \gamma(0)$ by definition. If genes are separated by non-coding spans, one adjoins an intergenic state IG to form the extended state space $\mathcal{C} \cup \{\mathrm{IG}\}$.
2. **Extended translation-recycling automaton:** For a linear mRNA translated by a recycling ribosome pool, we adjoin two boundary states—initiation $\perp_{\mathrm{start}}$ and termination $\top_{\mathrm{stop}}$—to form the extended state space $\widetilde{\mathcal{C}} := \mathcal{C}_{\mathrm{sense}} \cup \{\perp_{\mathrm{start}}, \top_{\mathrm{stop}}\}$. A complete translational cycle is a directed closed walk

$$\perp_{\mathrm{start}} \longrightarrow C_1 \longrightarrow C_2 \longrightarrow \cdots \longrightarrow C_{n-2} \longrightarrow \top_{\mathrm{stop}} \longrightarrow \perp_{\mathrm{start}},$$

   where the regeneration edge $\top_{\mathrm{stop}} \to \perp_{\mathrm{start}}$ represents post-termination ribosome release and 40S/60S subunit recycling back to the $5'$ initiation complex.

Proposition 3.5 (Irreversibility of the translation-recycling cycle)**.** *In the extended translation automaton on $\widetilde{\mathcal{C}}$, if ribosomal elongation and recycling proceed unidirectionally along a cycle $\gamma_{\mathrm{trans}} = (\perp_{\mathrm{start}}, c_1, \ldots, c_{n-2}, \top_{\mathrm{stop}}, \perp_{\mathrm{start}})$ with positive forward transition probabilities and strictly smaller (or zero) reverse transition probabilities, then the cycle affinity $\mathcal{A}(\gamma_{\mathrm{trans}}) > 0$ and the steady-state entropy production rate is strictly positive ($\sigma_{\mathrm{cycle}} > 0$).*

Proof. Along γ_{trans}, each forward transition probability exceeds its reverse counterpart, so the product of forward probabilities strictly exceeds the product of reverse probabilities. Taking the logarithm gives $\mathcal{A}(\gamma_{\mathrm{trans}}) > 0$, which by the cycle-affinity criterion proved in Theorem 8.1(iii) below implies $\sigma_{\mathrm{cycle}} > 0$. □

4 The genetic code as a fibration of the codon torus

Let $\mathcal{A}$ denote the set of functional translation channels of the genetic code. In the standard nuclear genetic code (NCBI Translation Table 1) [6, 7],

$$|\mathcal{A}| = 21,$$

comprising 20 canonical amino acids ($\mathcal{A}_{\text{sense}}$, $|\mathcal{A}_{\text{sense}}| = 20$) and one translation termination channel (Stop $= \{\text{TAA}, \text{TAG}, \text{TGA}\}$). The genetic code is a surjective map

$$\alpha : \mathbb{T}_4^3 \twoheadrightarrow \mathcal{A}.$$

For each channel $a \in \mathcal{A}$, define the synonymous fiber $F_a := \alpha^{-1}(a) \subset \mathbb{T}_4^3$ and its degeneracy (cardinality) $k_a := |F_a| \geq 1$.

Degeneracy class (k_a)	Channels $a \in \mathcal{A}$ in standard genetic code	Number of channels m_k	Total codons $m_k \cdot k$	Simplex dim $m_k(k-1)$
Singlet ($k_a = 1$)	Met, Trp	2	2	0
Doublet ($k_a = 2$)	Phe, Tyr, His, Gln, Asn, Lys, Asp, Glu, Cys	9	18	9
Triplet ($k_a = 3$)	Ile, Stop	2	6	4
Quartet ($k_a = 4$)	Val, Pro, Thr, Ala, Gly	5	20	15
Sextet ($k_a = 6$)	Leu, Ser, Arg	3	18	15
Total ($64 \to 21$)	**All 21 channels (20 AA + Stop)**	**21**	**64**	**43**
Sense ($61 \to 20$)	**20 sense amino-acid channels**	**20**	**61**	**41**

Table 1: Exact fiber multiplicity partition and synonymous simplex dimensions of the standard genetic code $\alpha : \mathbb{T}_4^3 \to \mathcal{A}$.

Proposition 4.1 (Exact fiber counting and synonymous simplex dimension)**.** *For each channel $a \in \mathcal{A}$, let*

$$\Delta_a := \left\{ p \in \mathbb{R}_{\geq 0}^{F_a} : \sum_{c \in F_a} p(c) = 1 \right\}$$

denote the probability simplex supported on the synonymous fiber F_a, and let $\Delta_{\text{syn}} := \prod_{a \in \mathcal{A}} \Delta_a$ denote the space of all conditional synonymous codon usage distributions $\{P(C \mid A = a)\}_{a \in \mathcal{A}}$. Then:

(i) The fiber cardinalities satisfy $\sum_{a \in \mathcal{A}} k_a = |\mathbb{T}_4^3| = 64$.

(ii) Each fiber simplex Δ_a is a compact convex polytope of real dimension $\dim_{\mathbb{R}} \Delta_a = k_a - 1$, and the product manifold-with-corners Δ_{syn} has exact real dimension

$$\dim_{\mathbb{R}} \Delta_{\text{syn}} = \sum_{a \in \mathcal{A}} (k_a - 1) = \sum_{a \in \mathcal{A}} k_a - |\mathcal{A}| = 64 - 21 = 43.$$

(iii) Restricted to the 61 sense codons and 20 amino-acid channels ($\mathcal{A}_{\text{sense}}$), the sense synonymous distribution space $\Delta_{\text{syn}}^{\text{sense}} = \prod_{a \in \mathcal{A}_{\text{sense}}} \Delta_a$ has exact real dimension

$$\dim_{\mathbb{R}} \Delta_{\text{syn}}^{\text{sense}} = 61 - 20 = 41.$$

(iv) For any fixed marginal amino-acid distribution $P(A)$ on $\mathcal{A}$,

$$\max_{P(C|A) \in \Delta_{\text{syn}}} H(C \mid A) = \sum_{a \in \mathcal{A}} P(A = a) \log k_a \leq \log 6 \approx 1.7918 \text{ nats } (2.5850 \text{ bits}).$$

Equality in the first identity holds if and only if $P(C = c \mid A = a) = 1/k_a$ is uniform on F_a for every channel a with $P(A = a) > 0$. Equality in the universal upper bound $\log 6$ holds if and only if $P(A)$ is supported exclusively on sextet channels ($k_a = 6$) and $P(C \mid A = a)$ is uniform on those fibers.

Proof. Since $\alpha : \mathbb{T}_4^3 \to \mathcal{A}$ is surjective, $\{F_a\}_{a\in\mathcal{A}}$ is a partition of $\mathbb{T}_4^3$ into $|\mathcal{A}| = 21$ non-empty subsets, giving (i). The affine hyperplane $\sum_{c\in F_a} p(c) = 1$ in $\mathbb{R}^{k_a}$ has dimension $k_a - 1$, and the non-negativity constraints $p(c) \geq 0$ cut out a simplex with non-empty relative interior of dimension $k_a - 1$. Since the dimension of a finite Cartesian product of manifolds-with-corners is the sum of the component dimensions, $\dim_{\mathbb{R}} \Delta_{\text{syn}} = \sum_{a\in\mathcal{A}}(k_a - 1) = 64 - 21 = 43$ (Table 1), proving (ii), and omitting the 3-codon Stop fiber ($k_{\text{Stop}} - 1 = 2$) gives $43 - 2 = 41$ in (iii). Finally, $H(C \mid A = a) = -\sum_{c\in F_a} P(c \mid a) \log P(c \mid a) \leq \log k_a$ by Jensen's inequality, with equality if and only if $P(c \mid a) = 1/k_a$, and since $k_a \leq 6$ for all $a \in \mathcal{A}$, $\sum_a P(A = a) \log k_a \leq \log 6$ with equality if and only if $P(A = a) = 0$ whenever $k_a < 6$, yielding (iv). □

Remark 4.2 (Code-agnostic formulas for alternative translation tables)**.** All dimension and degree-of-freedom formulas in this paper are code-agnostic. For any surjective genetic code $\alpha : \mathcal{C} \twoheadrightarrow \mathcal{A}$ (including vertebrate mitochondrial Table 2, yeast mitochondrial Table 3, *Mycoplasma*/protozoan Table 4, ciliate nuclear Table 6, or alternative yeast nuclear Table 12), the total synonymous simplex dimension is

$$\dim_{\mathbb{R}} \Delta_{\text{syn}} = \sum_{a\in\mathcal{A}} (k_a - 1) = |\mathcal{C}| - |\mathcal{A}|,$$

and the sense-only synonymous dimension on $\mathcal{C}_{\text{sense}} = \mathcal{C} \setminus \alpha^{-1}(\text{Stop})$ is $|\mathcal{C}_{\text{sense}}| - |\mathcal{A}_{\text{sense}}|$ (for example, in vertebrate mitochondrial Table 2, where AGA and AGG are additional stop codons so $|\alpha^{-1}(\text{Stop})| = 4$ and $|\mathcal{C}_{\text{sense}}| = 60$, the sense synonymous dimension is $60 - 20 = 40$; in *Mycoplasma* Table 4, where TGA is reassigned to Trp so $|\alpha^{-1}(\text{Stop})| = 2$ and $|\mathcal{C}_{\text{sense}}| = 62$, the sense synonymous dimension is $62 - 20 = 42$).

4.1 The two-strand duplex genetic-code fibration and the 8-doublet reduction

On a double-stranded DNA molecule, each sense codon $c = (c_1, c_2, c_3) \in \mathbb{T}_4^3$ is paired physically with its antiparallel Watson–Crick partner $\rho(c) = (\kappa(c_3), \kappa(c_2), \kappa(c_1))$. Evaluating the genetic code α on both strands of the 128-codon duplex defines the *two-strand (duplex) genetic-code map* [18, 22]

$$\alpha_{\text{dup}} : \mathbb{T}_4^3 \longrightarrow \mathcal{A} \times \mathcal{A}, \qquad \alpha_{\text{dup}}(c) := \big(\alpha(c),\, \alpha(\rho(c))\big) =: \big(A(c),\, A^{\rho}(c)\big).$$

For each ordered pair of channels $(a, b) \in \mathcal{A} \times \mathcal{A}$, let $F_{a,b} := \alpha_{\text{dup}}^{-1}(a, b) = F_a \cap \rho^{-1}(F_b)$ denote the joint sense–antisense fiber.

Theorem 4.3 (Two-strand duplex fibration, 8-doublet reduction, and synonymous entropy split)**.** *Under the standard genetic code $\alpha : \mathbb{T}_4^3 \to \mathcal{A}$ (NCBI Translation Table 1):*

(i) ***Wobble-to-leading-base coupling and fiber sizes:*** *Because $\rho(c_1, c_2, c_3) = (\kappa(c_3), \kappa(c_2), \kappa(c_1))$ maps the third (wobble) position c_3 of the sense codon onto the first (leading) position $\kappa(c_3)$ of the antisense codon, two distinct codons $c \neq c'$ share the same joint channel $\alpha_{\text{dup}}(c) = \alpha_{\text{dup}}(c')$ if and only if they belong to one of exactly* ***8 binary doublets*** *($|F_{a,b}| = 2$), comprising 4 pairs whose reverse complements lie in the two-box families of Leucine or Arginine and 4 dual pairs in Leucine or Arginine:*

$$F_{\text{Stop,Leu}} = \{\text{TAA}, \text{TAG}\}, \quad F_{\text{Gln,Leu}} = \{\text{CAA}, \text{CAG}\}, \quad F_{\text{Ser,Arg}} = \{\text{TCG}, \text{TCT}\}, \quad F_{\text{Pro,Arg}} = \{\text{CCG}, \text{CCT}$$
$$F_{\text{Leu,Stop}} = \{\text{TTA}, \text{CTA}\}, \quad F_{\text{Leu,Gln}} = \{\text{TTG}, \text{CTG}\}, \quad F_{\text{Arg,Ser}} = \{\text{CGA}, \text{AGA}\}, \quad F_{\text{Arg,Pro}} = \{\text{CGG}, \text{AGG}$$

The remaining $64 - 16 = 48$ codons are ***singletons*** *($|F_{a,b}| = 1$), uniquely determined by their sense–antisense channel pair $(A(c), A^{\rho}(c))$.*

(ii) ***Reduction of synonymous dimension from 43 to 8:*** *The image $\text{Im}(\alpha_{\text{dup}}) \subset \mathcal{A} \times \mathcal{A}$ consists of exactly $48 + 8 = 56$ joint channels (or 54 on the 61 sense codons), and the joint duplex synonymous simplex space $\Delta_{\text{dup}} := \prod_{(a,b)\in\text{Im}(\alpha_{\text{dup}})} \Delta_{F_{a,b}}$ has exact real dimension*

$$\dim_{\mathbb{R}} \Delta_{\text{dup}} = \sum_{(a,b)\in\text{Im}(\alpha_{\text{dup}})} \big(|F_{a,b}| - 1\big) = 64 - 56 = 8$$

(7 when restricted to the 61 sense codons, and 6 when both strands are restricted to sense codons).

(iii) ***Exact two-strand decomposition of single-strand synonymous entropy:*** *For any codon random variable C on $\mathbb{T}_4^3$ with $A = \alpha(C)$ and $A^\rho = \alpha(\rho(C))$, the 43-dimensional single-strand synonymous conditional entropy decomposes as*

$$H(C \mid A) = H(A^\rho \mid A) + H(C \mid A, A^\rho), \qquad 0 \le H(C \mid A, A^\rho) \le P(C \in \mathcal{D}_8) \log 2, \quad (1)$$

where $\mathcal{D}_8$ is the union of the 8 binary doublets ($|\mathcal{D}_8| = 16$).

Proof. (i)–(ii) If $c \neq c'$ satisfy $\alpha(c) = \alpha(c') = a$ and $\alpha(\rho(c)) = \alpha(\rho(c')) = b$, then $\rho(c) \neq \rho(c')$ are synonymous in F_b. In Table 1, synonymous codons within a fiber differ either at the 3rd base only (all 2-, 3-, and 4-codon fibers, plus the 4-box and 2-box subfamilies of Leu, Ser, Arg), or at the 1st base with identical 2nd and 3rd bases (precisely the four inter-box pairs $\{\mathrm{TTA}, \mathrm{CTA}\}$ and $\{\mathrm{TTG}, \mathrm{CTG}\}$ in Leu, and $\{\mathrm{CGA}, \mathrm{AGA}\}$ and $\{\mathrm{CGG}, \mathrm{AGG}\}$ in Arg), or at both 1st and 2nd bases (Ser TCN vs. AGT/AGC) or 2nd and 3rd bases (Stop TAG vs. TGA). Under ρ, a 3rd-base difference in c, c' becomes a 1st-base difference in $\rho(c), \rho(c')$ with identical 2nd and 3rd bases; hence $\rho(c), \rho(c')$ are synonymous if and only if $\{\rho(c), \rho(c')\}$ is one of the four Leu/Arg inter-box pairs above, which gives the first four doublets $\{\mathrm{TAA}, \mathrm{TAG}\}$, $\{\mathrm{CAA}, \mathrm{CAG}\}$, $\{\mathrm{TCG}, \mathrm{TCT}\}$, $\{\mathrm{CCG}, \mathrm{CCT}\}$, and applying ρ gives the four dual doublets. Direct inspection of Ser TCN vs. AGT/AGC (ρ-images have channels $\{\mathrm{Arg}, \mathrm{Gly}, \mathrm{Stop}\}$ vs. $\{\mathrm{Thr}, \mathrm{Ala}\}$, disjoint) and Stop TGA ($\rho(\mathrm{TGA}) = \mathrm{TCA} \in F_{\mathrm{Ser}} \neq F_{\mathrm{Leu}}$) confirms that no other pairs exist. Thus there are 8 doublets (16 codons) and 48 singletons, giving $|\mathrm{Im}(\alpha_{\mathrm{dup}})| = 56$ and $\dim_{\mathbb{R}} \Delta_{\mathrm{dup}} = 64 - 56 = 8$. (iii) Since $A^\rho = \alpha(\rho(C))$ is a deterministic function of C, $H(C, A^\rho \mid A) = H(C \mid A)$. Applying the chain rule $H(C, A^\rho \mid A) = H(A^\rho \mid A) + H(C \mid A, A^\rho)$ yields (1), and since $|F_{a,b}| = 1$ outside $\mathcal{D}_8$ and $|F_{a,b}| = 2$ on $\mathcal{D}_8$, $H(C \mid A = a, A^\rho = b) \le \log 2$ on $\mathcal{D}_8$ and 0 elsewhere. □

Corollary 4.4 (Antisense termination and initiation preimages and dual frame-opening under wobble-A/T depletion)**.** *Under the standard nuclear genetic code $\alpha : \mathbb{T}_4^3 \to \mathcal{A}$ (Table 1), let $\gamma = (c_1, \ldots, c_n) \in (\mathbb{T}_4^3)^n$ be a framed sense coding sequence of length n codons ($3n$ nucleotides). Then:*

(i) ***Antisense termination and initiation preimages in frame -0:*** *Under the reverse-complement involution $\rho(c) = -Rc + \mathbf{31} \pmod 4$, the preimage of the termination channel* $\mathrm{Stop} = \{\mathrm{TAA}, \mathrm{TAG}, \mathrm{TGA}\}$ *is the 3-codon union of the binary doublet $F_{\mathrm{Leu,Stop}} = \{\mathrm{TTA}, \mathrm{CTA}\}$ and the singleton $F_{\mathrm{Ser,Stop}} = \{\mathrm{TCA}\}$, whereas the preimage of the canonical initiation codon* ATG *is the Histidine singleton $F_{\mathrm{His,Met}} = \{\mathrm{CAT}\}$:*

$$\rho^{-1}\big(\alpha^{-1}(\mathrm{Stop})\big) = F_{\mathrm{Leu,Stop}} \cup F_{\mathrm{Ser,Stop}} = \{\mathrm{TTA}, \mathrm{CTA}, \mathrm{TCA}\}, \qquad \rho^{-1}(\{\mathrm{ATG}\}) = \{\mathrm{CAT}\}.$$

Every codon in $\rho^{-1}(\alpha^{-1}(\mathrm{Stop}))$ has adenine at the third (wobble) position ($c_{i,3} = A$, $\mathrm{GC3} = 0$), and CAT *has thymine at the third position ($c_{i,3} = T$, $\mathrm{GC3} = 0$).*

(ii) ***Frame-$+2$ termination condition on the sense strand:*** *In the shifted sense reading frame $+2$, the i-th triplet is $(c_{i,3}, c_{i+1,1}, c_{i+1,2})$ ($1 \le i \le n-1$). Since every stop codon in* $\mathrm{Stop} = \{\mathrm{TAA}, \mathrm{TAG}, \mathrm{TGA}\}$ *begins with T, a frame-$+2$ triplet can be a stop codon only if the preceding sense codon ends in thymine ($c_{i,3} = T$, $\mathrm{GC3} = 0$).*

(iii) ***Dual frame-opening and* CAT*-initiated antisense open reading frames:*** *Any recoding policy that eliminates wobble-A and wobble-T sense codons ($\mathrm{GC3} = 1$, or human deterministic Max-CAI where $\phi(\mathrm{Leu}) = \mathrm{CTG}$, $\phi(\mathrm{Ser}) = \mathrm{AGC}$, and $\phi(\mathrm{His}) = \mathrm{CAC}$) simultaneously drives the number of in-frame termination codons in antisense frame -0 to zero ($\#\{i : \alpha(\rho(c_i)) = \mathrm{Stop}\} = 0$), drives the number of termination codons in sense frame $+2$ to zero, and eliminates antisense frame--0* ATG *start codons ($\#\{i : c_i = \mathrm{CAT}\} = 0$). Conversely, any codon-optimization protocol that depletes* $\{\mathrm{TTA}, \mathrm{CTA}, \mathrm{TCA}\}$ *to zero while retaining sense* CAT *codons simultaneously removes all termination barriers in antisense frame -0 and supplies in-frame antisense* $\mathrm{ATG} = \rho(\mathrm{CAT})$ *initiation sites, generating a continuous*

antiparallel ATG $\to$ End *open reading frame spanning from the last sense* CAT *codon to the* $5'$ *boundary of the sense gene.*

Proof. Part (i) follows directly from Proposition 3.3 and Theorem 4.3(i): ρ is an involution on $\mathbb{T}_4^3$ with $\rho(\mathrm{TAA}) = \mathrm{TTA} \in F_{\mathrm{Leu,Stop}}$, $\rho(\mathrm{TAG}) = \mathrm{CTA} \in F_{\mathrm{Leu,Stop}}$, $\rho(\mathrm{TGA}) = \mathrm{TCA} \in F_{\mathrm{Ser,Stop}}$, and $\rho(\mathrm{ATG}) = \mathrm{CAT} \in F_{\mathrm{His,Met}}$. Part (ii) holds because the first coordinate of the frame-$+2$ triplet $(c_{i,3}, c_{i+1,1}, c_{i+1,2})$ is $c_{i,3}$, and $\alpha^{-1}(\mathrm{Stop}) \subset \{T\} \times B^2$. Part (iii) follows immediately from (i) and (ii): if $\{c_i\} \cap \{\mathrm{TTA}, \mathrm{CTA}, \mathrm{TCA}\} = \varnothing$, then $\alpha(\rho(c_i)) \neq \mathrm{Stop}$ for all $i = 1, \ldots, n$; if $c_{i,3} \neq T$ for all i, no frame-$+2$ triplet can start with T and no c_i can equal CAT; whereas if $c_{i_0} = \mathrm{CAT}$ for some index i_0 while $\{c_1, \ldots, c_{i_0}\} \cap \{\mathrm{TTA}, \mathrm{CTA}, \mathrm{TCA}\} = \varnothing$, then the antiparallel frame--0 trajectory $\big(\rho(c_{i_0}), \rho(c_{i_0-1}), \ldots, \rho(c_1)\big)$ begins with $\rho(\mathrm{CAT}) = \mathrm{ATG}$ and contains zero stop codons across all i_0 triplets. □

5 Synonymous program information, chain-rule decompositions, and Markov lumpability

The following information-theoretic quantities follow Shannon's foundational definitions [1, 2]. Let (C_{i-1}, C_i) be any pair of consecutive codons taking values in $\mathcal{C} \times \mathcal{C}$, and let $A_{i-1} := \alpha(C_{i-1}) \in \mathcal{A}$ and $A_i := \alpha(C_i) \in \mathcal{A}$ be the corresponding functional channels.

Definition 5.1 (Contextual synonymous program information)**.** The *contextual synonymous program information* is the conditional mutual information between consecutive codons given the current functional channel:

$$S_{\mathrm{syn}} := I(C_{i-1}; C_i \mid A_i).$$

Theorem 5.2 (Non-negativity and characterization of S_{syn})**.** *For any joint distribution of* (C_{i-1}, C_i) *with* $A_i = \alpha(C_i)$*,*

$$S_{\mathrm{syn}} = I(C_{i-1}; C_i \mid A_i) = \mathbb{E}_{A_i, C_{i-1}}\Big[D_{\mathrm{KL}}\big(P(C_i \mid C_{i-1}, A_i) \,\big\|\, P(C_i \mid A_i)\big)\Big] \geq 0,$$

with $S_{\mathrm{syn}} = 0$ *if and only if* C_{i-1} *and* C_i *are conditionally independent given* A_i *(*$C_{i-1} \perp C_i \mid A_i$*). Moreover,* S_{syn} *is bounded above by the marginal synonymous entropy:*

$$0 \leq S_{\mathrm{syn}} \leq H(C_i \mid A_i) \leq \sum_{a \in \mathcal{A}} P(A_i = a) \log k_a.$$

Proof. By definition of conditional mutual information [1, 2],

$$I(C_{i-1}; C_i \mid A_i) = \sum_{a \in \mathcal{A}} \sum_{x \in \mathcal{C}} P(C_{i-1} = x, A_i = a) \times \sum_{y \in F_a} P(C_i = y \mid C_{i-1} = x, A_i = a) \log \frac{P(C_i = y \mid C_{i-1} = x, A_i = a)}{P(C_i = y \mid A_i = a)}.$$

The inner sum over $y \in F_a$ is $D_{\mathrm{KL}}\big(P(C_i \mid C_{i-1} = x, A_i = a) \,\|\, P(C_i \mid A_i = a)\big) \geq 0$, which vanishes if and only if $P(C_i = y \mid C_{i-1} = x, A_i = a) = P(C_i = y \mid A_i = a)$ for all $y \in F_a$. Finally, $I(C_{i-1}; C_i \mid A_i) = H(C_i \mid A_i) - H(C_i \mid C_{i-1}, A_i) \leq H(C_i \mid A_i)$, and Proposition 4.1(iv) gives the upper bound. □

5.1 Tripartite and quadripartite chain-rule decompositions

We now prove that the total sequential mutual information $I(C_{i-1}; C_i)$ on the codon torus decomposes into three (or four) additive, non-negative chain-rule components.

Theorem 5.3 (Tripartite decomposition of codon-bigram mutual information)**.** *For any joint distribution of two consecutive codons (C_{i-1}, C_i) with $A_{i-1} = \alpha(C_{i-1})$ and $A_i = \alpha(C_i)$, with no stationarity assumption required, the total codon-bigram mutual information $I(C_{i-1}; C_i)$ decomposes into the sum of three additive non-negative chain-rule components:*

$$I(C_{i-1}; C_i) = \underbrace{I(A_{i-1}; A_i)}_{\text{(I) Protein dipeptide syntax}} + \underbrace{I(C_{i-1}; A_i \mid A_{i-1})}_{\text{(II) Codon-to-next-AA context}} + \underbrace{I(C_{i-1}; C_i \mid A_i)}_{\text{(III) Synonymous program } S_{\text{syn}}}. \quad (2)$$

Furthermore, let $P_{\text{emp}}(c_i \mid c_{i-1}) = P(C_i = c_i \mid C_{i-1} = c_{i-1})$ be the true bigram transition kernel, and consider the three standard reference null kernels [10]:

$$\begin{aligned}
q_0(c_i \mid c_{i-1}) &:= P(C_i = c_i) && \text{(H0: marginal codon usage)},\\
q_2(c_i \mid c_{i-1}) &:= P(A_i = a_i \mid A_{i-1} = a_{i-1})\, P(C_i = c_i \mid A_i = a_i) && \text{(H2: AA bigram + marg. syn.)},\\
q_{2b}(c_i \mid c_{i-1}) &:= P(A_i = a_i \mid A_{i-1} = a_{i-1})\, P(C_i = c_i \mid C_{i-1} = c_{i-1}, A_i = a_i) && \text{(H2b: AA bigram + cond. syn.)}.
\end{aligned}$$

Denoting the expected log-likelihood excesses by $d_k := \mathbb{E}\big[\log\big(P_{\text{emp}}(C_i \mid C_{i-1})/q_k(C_i \mid C_{i-1})\big)\big]$ for $k \in \{0, 2, 2b\}$ whenever the kernels are strictly positive on the support of the true distribution (or using the standard extended-real convention and Dirichlet smoothing in practice), we have the exact identities:

$$d_0 = I(C_{i-1}; C_i), \quad d_0 - d_2 = I(A_{i-1}; A_i), \quad d_{2b} = I(C_{i-1}; A_i \mid A_{i-1}), \quad d_2 - d_{2b} = S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i). \quad (3)$$

Proof. Since $A_i = \alpha(C_i)$ is a deterministic function of C_i, $H(A_i \mid C_i) = H(A_i \mid C_{i-1}, C_i) = 0$, so $I(C_{i-1}; C_i) = I(C_{i-1}; C_i, A_i)$. By the chain rule for mutual information,

$$I(C_{i-1}; C_i, A_i) = I(C_{i-1}; A_i) + I(C_{i-1}; C_i \mid A_i).$$

Similarly, since $A_{i-1} = \alpha(C_{i-1})$ is a deterministic function of C_{i-1}, $I(C_{i-1}; A_i) = I(C_{i-1}, A_{i-1}; A_i)$. Expanding this first term via the chain rule yields

$$I(C_{i-1}, A_{i-1}; A_i) = I(A_{i-1}; A_i) + I(C_{i-1}; A_i \mid A_{i-1}).$$

Combining the two expansions establishes (2).

To prove (3), observe that for any $c_{i-1}, c_i \in \mathcal{C}$ with $a_{i-1} = \alpha(c_{i-1})$ and $a_i = \alpha(c_i)$, the exact conditional probability factors as

$$P_{\text{emp}}(c_i \mid c_{i-1}) = P(A_i = a_i \mid C_{i-1} = c_{i-1})\, P(C_i = c_i \mid C_{i-1} = c_{i-1}, A_i = a_i).$$

Dividing $P_{\text{emp}}(c_i \mid c_{i-1})$ by $q_{2b}(c_i \mid c_{i-1})$ cancels the second factor identically, leaving

$$\frac{P_{\text{emp}}(c_i \mid c_{i-1})}{q_{2b}(c_i \mid c_{i-1})} = \frac{P(A_i = a_i \mid C_{i-1} = c_{i-1}, A_{i-1} = a_{i-1})}{P(A_i = a_i \mid A_{i-1} = a_{i-1})}.$$

Taking the expectation over (C_{i-1}, C_i) yields $d_{2b} = I(C_{i-1}; A_i \mid A_{i-1})$. Likewise, dividing $q_{2b}(c_i \mid c_{i-1})$ by $q_2(c_i \mid c_{i-1})$ cancels $P(A_i = a_i \mid A_{i-1} = a_{i-1})$, giving

$$d_2 - d_{2b} = \mathbb{E}\left[\log \frac{q_{2b}(C_i \mid C_{i-1})}{q_2(C_i \mid C_{i-1})}\right] = \mathbb{E}\left[\log \frac{P(C_i \mid C_{i-1}, A_i)}{P(C_i \mid A_i)}\right] = I(C_{i-1}; C_i \mid A_i) = S_{\text{syn}}.$$

Finally, $d_0 = \mathbb{E}[\log(P(C_i \mid C_{i-1})/P(C_i))] = I(C_{i-1}; C_i)$, and subtracting $d_2 = d_{2b} + S_{\text{syn}}$ from (2) gives $d_0 - d_2 = I(A_{i-1}; A_i)$. □

Theorem 5.4 (Quadripartite decomposition and strictly pairwise synonymous residual)**.** *The contextual synonymous information $S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i)$ conditions on the current amino-acid*

channel A_i, and therefore decomposes further by conditioning on the preceding amino-acid channel A_{i-1}:

$$S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i) = I(A_{i-1}; C_i \mid A_i) + S^{(2)}_{\text{syn}}, \qquad S^{(2)}_{\text{syn}} := I(C_{i-1}; C_i \mid A_{i-1}, A_i) \geq 0.$$

Consequently, the total codon-bigram mutual information admits the exact four-term decomposition:

$$I(C_{i-1}; C_i) = \underbrace{I(A_{i-1}; A_i)}_{AA \to AA \text{ syntax}} + \underbrace{I(C_{i-1}; A_i \mid A_{i-1})}_{Syn_1 \to AA_2 \text{ context}} + \underbrace{I(A_{i-1}; C_i \mid A_i)}_{AA_1 \to Syn_2 \text{ context}} + \underbrace{I(C_{i-1}; C_i \mid A_{i-1}, A_i)}_{\text{Purely synonymous } S^{(2)}_{\text{syn}}}. \quad (4)$$

Moreover, defining the amino-acid-pair-conditioned synonymous null kernel

$$q_{2,\text{pair}}(c_i \mid c_{i-1}) := P(A_i = a_i \mid A_{i-1} = a_{i-1})\, P(C_i = c_i \mid A_{i-1} = a_{i-1}, A_i = a_i)$$

and its expected log-likelihood excess $d_{2,\text{pair}} := \mathbb{E}\big[\log\big(P_{\text{emp}}(C_i \mid C_{i-1})/q_{2,\text{pair}}(C_i \mid C_{i-1})\big)\big]$, we have the exact identities:

$$d_2 - d_{2,\text{pair}} = I(A_{i-1}; C_i \mid A_i), \qquad d_{2,\text{pair}} - d_{2b} = S^{(2)}_{\text{syn}} = I(C_{i-1}; C_i \mid A_{i-1}, A_i).$$

Proof. Since $A_{i-1} = \alpha(C_{i-1})$ is a deterministic function of C_{i-1}, $I(C_{i-1}; C_i \mid A_i) = I(C_{i-1}, A_{i-1}; C_i \mid A_i)$. Applying the chain rule for conditional mutual information gives

$$I(C_{i-1}, A_{i-1}; C_i \mid A_i) = I(A_{i-1}; C_i \mid A_i) + I(C_{i-1}; C_i \mid A_{i-1}, A_i).$$

Substituting this into (2) yields (4). Furthermore, because $A_{i-1} = \alpha(C_{i-1})$ is determined by C_{i-1},

$$P(C_i = c_i \mid C_{i-1} = c_{i-1}, A_i = a_i) = P(C_i = c_i \mid C_{i-1} = c_{i-1}, A_{i-1} = a_{i-1}, A_i = a_i),$$

whence

$$d_{2,\text{pair}} - d_{2b} = \mathbb{E}\left[\log \frac{q_{2b}(C_i \mid C_{i-1})}{q_{2,\text{pair}}(C_i \mid C_{i-1})}\right] = \mathbb{E}\left[\log \frac{P(C_i \mid C_{i-1}, A_{i-1}, A_i)}{P(C_i \mid A_{i-1}, A_i)}\right] = I(C_{i-1}; C_i \mid A_{i-1}, A_i) = S^{(2)}_{\text{syn}},$$

and subtracting from $d_2 - d_{2b} = S_{\text{syn}}$ gives $d_2 - d_{2,\text{pair}} = I(A_{i-1}; C_i \mid A_i)$. □

Theorem 5.5 (Duplex antiparallel invariance and two-strand decomposition of S_{syn})**.** *Let (C_{i-1}, C_i) be any pair of consecutive codons on the sense strand, let $(\widetilde{C}_{i-1}, \widetilde{C}_i) := (\rho(C_i), \rho(C_{i-1}))$ be the corresponding antiparallel $5' \to 3'$ transition on the complementary strand, and let $A_i = \alpha(C_i)$ and $A^\rho_i = \alpha(\rho(C_i))$. Then:*

(i) ***Duplex invariance of total bigram mutual information:***

$$I(\widetilde{C}_{i-1}; \widetilde{C}_i) = I\big(\rho(C_i); \rho(C_{i-1})\big) = I(C_{i-1}; C_i).$$

(ii) ***Exact two-strand split of the synonymous program:*** *The contextual synonymous program information $S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i)$ decomposes into the sum of two non-negative two-strand components:*

$$S_{\text{syn}} = \underbrace{I(C_{i-1};\, A^\rho_i \mid A_i)}_{S_{\text{anti}} \text{ (antisense } AA \text{ program)}} + \underbrace{I(C_{i-1};\, C_i \mid A_i, A^\rho_i)}_{S_{\text{dup}} \text{ (8-doublet residual)}}, \quad (5)$$

where $S_{\text{anti}} \geq 0$ measures the predictive information that the preceding codon C_{i-1} carries about the antisense amino-acid channel A^ρ_i given A_i, and $S_{\text{dup}} \geq 0$ is supported exclusively on the 8 binary doublets of Theorem 4.3.

Proof. (i) Because $\rho : \mathbb{T}^3_4 \to \mathbb{T}^3_4$ is a bijection and mutual information is invariant under bijections of its arguments and symmetric in (X, Y), $I(\rho(C_i); \rho(C_{i-1})) = I(C_i; C_{i-1}) = I(C_{i-1}; C_i)$. (ii) Since $A^\rho_i = \alpha(\rho(C_i))$ is a deterministic function of C_i, $H(A^\rho_i \mid C_i, A_i) = H(A^\rho_i \mid C_{i-1}, C_i, A_i) = 0$, so $I(C_{i-1}; C_i \mid A_i) = I(C_{i-1}; C_i, A^\rho_i \mid A_i)$. Expanding the right-hand side by the chain rule for conditional mutual information gives (5). Whenever (A_i, A^ρ_i) lies in one of the 48 singleton fibers of Theorem 4.3, C_i is uniquely determined by (A_i, A^ρ_i), so those strata contribute zero to S_{dup}. □

5.2 Markov lumpability of the genetic code projection

When is the projected amino-acid sequence $A_i = \alpha(C_i)$ an autonomous first-order Markov chain whose transition probabilities depend only on the preceding amino acid A_{i-1}? In the theory of finite Markov chains, this is the question of *strong lumpability* of the transition matrix P with respect to the partition $\{F_a\}_{a\in\mathcal{A}}$ [8].

Theorem 5.6 (Information-theoretic criterion for Markov lumpability of A_i)**.** *Let (C_i) be a first-order Markov chain on $\mathbb{T}_4^3$ with transition matrix $P_{xy} = P(C_i = y \mid C_{i-1} = x)$ and strictly positive marginal codon probabilities $P(C_{i-1} = x) > 0$. The following two conditions are equivalent:*

(i) ***Kemeny–Snell lumpability:*** *For every pair of channels $a, b \in \mathcal{A}$ and all synonymous codons $x, x' \in F_a$, the total transition probability into fiber F_b is independent of the synonymous choice in F_a:*

$$\sum_{y\in F_b} P_{xy} = \sum_{y\in F_b} P_{x'y}.$$

(ii) ***Zero codon-to-next-amino-acid context*** *($d_{2b} = 0$)**:***

$$I(C_{i-1}; A_i \mid A_{i-1}) = d_{2b} = 0.$$

Whenever $d_{2b} = I(C_{i-1}; A_i \mid A_{i-1}) > 0$, the amino-acid sequence is not a lumped Markov chain of the codon process: the synonymous choice of C_{i-1} carries strictly positive predictive information about the next amino acid A_i.

Proof. For $x \in F_a$, $\sum_{y\in F_b} P_{xy} = P(A_i = b \mid C_{i-1} = x, A_{i-1} = a)$. Condition (i) holds for all $x, x' \in F_a$ if and only if $P(A_i = b \mid C_{i-1} = x, A_{i-1} = a) = P(A_i = b \mid A_{i-1} = a)$ for all $a, b \in \mathcal{A}$ and $x \in F_a$. Since $P(C_{i-1} = x) > 0$, this conditional independence $C_{i-1} \perp A_i \mid A_{i-1}$ holds if and only if $I(C_{i-1}; A_i \mid A_{i-1}) = 0$, which equals d_{2b} by Theorem 5.3. □

6 Deterministic codon optimization collapses the synonymous program

In industrial gene synthesis and mRNA therapeutic design, standard "codon optimization" (such as maximizing the Codon Adaptation Index, Max-CAI [9]) selects a single preferred synonymous codon $\phi(a) \in F_a$ for each amino-acid channel $a \in \mathcal{A}$ and recodes the sequence deterministically as

$$C_i = \phi(A_i).$$

This amino-acid-deterministic recoding is distinct from, though related in motivation to, genome-scale codon-pair deoptimization strategies used in viral attenuation [11].

Theorem 6.1 (Total non-protein syntax collapse under deterministic Max-CAI designs)**.** *Suppose a coding sequence is generated by a deterministic section $\phi : \mathcal{A} \to \mathcal{C}$ of the genetic code ($\alpha \circ \phi = \mathrm{id}_{\mathcal{A}}$), so that $C_i = \phi(A_i)$ almost surely for all i. Then:*

(i) The conditional synonymous entropy vanishes: $H(C_i \mid A_i) = 0$.

(ii) All conditional non-dipeptide terms appearing in the tripartite and quadripartite decompositions (2) and (4) vanish identically:

$$S_{\mathrm{syn}} = I(C_{i-1}; C_i \mid A_i) = 0, \qquad S^{(2)}_{\mathrm{syn}} = I(C_{i-1}; C_i \mid A_{i-1}, A_i) = 0,$$
$$I(C_{i-1}; A_i \mid A_{i-1}) = 0, \qquad I(A_{i-1}; C_i \mid A_i) = 0.$$

(iii) Consequently, the entire codon-torus mutual information collapses to the bare amino-acid dipeptide syntax:

$$I(C_{i-1}; C_i) = I(A_{i-1}; A_i).$$

(iv) Every intra-gene synonymous multiset shuffle, which permutes synonymous codons at positions sharing the same amino acid A_i, leaves a Max-CAI sequence pointwise invariant. Consequently, any audit statistic based solely on reordering codons within identical amino-acid channels is identically zero on a deterministic Max-CAI design.

Proof. Since $C_i = \phi(A_i)$ is a deterministic function of A_i, $H(C_i \mid A_i) = 0$, $H(C_i \mid C_{i-1}, A_i) = 0$, and $H(C_i \mid A_{i-1}, A_i) = 0$, which implies $S_{\text{syn}} = 0$, $S^{(2)}_{\text{syn}} = 0$, and $I(A_{i-1}; C_i \mid A_i) = 0$. Likewise, at index $i-1$, $C_{i-1} = \phi(A_{i-1})$ is a deterministic function of A_{i-1}, so $H(C_{i-1} \mid A_{i-1}) = 0$ and $H(C_{i-1} \mid A_i, A_{i-1}) = 0$, whence

$$I(C_{i-1}; A_i \mid A_{i-1}) = H(C_{i-1} \mid A_{i-1}) - H(C_{i-1} \mid A_i, A_{i-1}) = 0 - 0 = 0,$$

proving (i) and (ii). Substituting (ii) into (2)–(4) gives (iii). Finally, under $C_i = \phi(A_i)$, all positions encoding amino acid a carry the identical codon $\phi(a)$, so any permutation of codons within the positions of channel a is the identity map, proving (iv). □

Proposition 6.2 (Cross-scoring penalty of Max-CAI under a natural reference kernel)**.** *Let P_{WT} be a wild-type reference Markov transition kernel on $\mathcal{C}$, and let a protein with amino-acid bigram distribution $P_{\text{prot}}(A_{i-1} = a, A_i = b)$ be recoded via $C_i = \phi(A_i)$. Assume that all conditioning events below have positive P_{WT}-probability; otherwise replace the empirical kernel by a Dirichlet-smoothed version, which leaves the sign conclusion unchanged on the support. When the recoded sequence is scored under the wild-type model P_{WT}, its expected log-likelihood excess over the wild-type H2b baseline $q_{2b,\text{WT}}$ is*

$$d_{2b}(\phi; P_{\text{WT}}) = \sum_{a,b \in \mathcal{A}} P_{\text{prot}}(A_{i-1} = a, A_i = b) \log \frac{P_{\text{WT}}\big(A_i = b \mid C_{i-1} = \phi(a)\big)}{P_{\text{WT}}(A_i = b \mid A_{i-1} = a)}. \tag{6}$$

In particular, if $P_{\text{prot}}(A_i = \cdot \mid A_{i-1} = a) = P_{\text{WT}}(A_i = \cdot \mid A_{i-1} = a)$, then

$$d_{2b}(\phi; P_{\text{WT}}) = -\sum_{a \in \mathcal{A}} P_{\text{prot}}(A_{i-1} = a)\, D_{\text{KL}}\Big(P_{\text{WT}}(A_i \mid A_{i-1} = a) \,\Big\|\, P_{\text{WT}}\big(A_i \mid C_{i-1} = \phi(a)\big)\Big) \le 0,$$

which is strictly negative whenever the single globally preferred codon $\phi(a)$ has a downstream amino-acid context distribution $P_{\text{WT}}(A_i \mid C_{i-1} = \phi(a))$ that differs from the channel average $P_{\text{WT}}(A_i \mid A_{i-1} = a)$.

Proof. As shown in the proof of Theorem 5.3, for any transition (c_{i-1}, c_i) with $\alpha(c_{i-1}) = a$ and $\alpha(c_i) = b$,

$$\log \frac{P_{\text{WT}}(c_i \mid c_{i-1})}{q_{2b,\text{WT}}(c_i \mid c_{i-1})} = \log \frac{P_{\text{WT}}(A_i = b \mid C_{i-1} = c_{i-1})}{P_{\text{WT}}(A_i = b \mid A_{i-1} = a)}.$$

Setting $c_{i-1} = \phi(a)$ and averaging over $P_{\text{prot}}(A_{i-1} = a, A_i = b)$ gives (6). When $P_{\text{prot}}(b \mid a) = P_{\text{WT}}(b \mid a)$, the inner sum over $b \in \mathcal{A}$ is precisely $-D_{\text{KL}}\big(P_{\text{WT}}(A_i \mid A_{i-1} = a) \,\|\, P_{\text{WT}}(A_i \mid C_{i-1} = \phi(a))\big) \le 0$. □

Proposition 6.3 (Variational optimality of first-order conditional synonymous compilation)**.** *Let P_{WT} be a wild-type transition kernel on $\mathcal{C}$, and consider the class of all first-order synonymous recoding policies $Q(C_i = y \mid C_{i-1} = x, A_i = a)$ supported on F_a. For any fixed distribution of (C_{i-1}, A_i), the expected wild-type conditional synonymous log-likelihood is bounded above by the wild-type conditional entropy:*

$$\mathbb{E}_Q\big[\log P_{\text{WT}}(C_i \mid C_{i-1}, A_i)\big] = -H_{P_{\text{WT}}}(C_i \mid C_{i-1}, A_i) - \mathbb{E}_{C_{i-1}, A_i}\Big[D_{\text{KL}}\big(Q(\cdot \mid C_{i-1}, A_i) \,\big\|\, P_{\text{WT}}(\cdot \mid C_{i-1}, A_i)\big)\Big],$$

achieving its unique maximum if and only if $Q(y \mid x, a) = P_{\text{WT}}(C_i = y \mid C_{i-1} = x, A_i = a)$ almost surely, in which case the compiled sequence preserves the full contextual synonymous program $S_{\text{syn}}(Q) = S_{\text{syn}}(P_{\text{WT}}) > 0$.

Proof. By definition of the Kullback–Leibler divergence on each fiber F_a,

$$\sum_{y\in F_a} Q(y \mid x,a)\log P_{\mathrm{WT}}(y \mid x,a) = \sum_{y\in F_a} Q(y \mid x,a)\log Q(y \mid x,a) - D_{\mathrm{KL}}\big(Q(\cdot \mid x,a) \,\|\, P_{\mathrm{WT}}(\cdot \mid x,a)\big).$$

At $Q = P_{\mathrm{WT}}(\cdot \mid x,a)$, the KL penalty vanishes identically and $S_{\mathrm{syn}}(Q) = I_{P_{\mathrm{WT}}}(C_{i-1}; C_i \mid A_i)$. □

7 Chemical data-processing bridge

Let $q : \mathcal{A} \to \mathcal{Q}$ be any deterministic coarsening of the functional channels into physicochemical classes (such as hydrophobic/polar/acidic/basic polarity classes, side-chain volume classes, secondary-structure propensity classes, or post-translational modification classes), and define $Q_i := q(A_i)$.

Theorem 7.1 (Chemical data-processing inequality and exact residual)**.** *For any codon process (C_i) with $A_i = \alpha(C_i)$ and $Q_i = q(A_i)$, the conditional mutual information between the preceding codon C_{i-1} and the next chemical class Q_i given A_{i-1} satisfies*

$$I(C_{i-1}; Q_i \mid A_{i-1}) \le I(C_{i-1}; A_i \mid A_{i-1}) = d_{2b}, \tag{7}$$

with the exact chain-rule decomposition

$$I(C_{i-1}; A_i \mid A_{i-1}) = I(C_{i-1}; Q_i \mid A_{i-1}) + I(C_{i-1}; A_i \mid Q_i, A_{i-1}). \tag{8}$$

Proof. Set $X = C_{i-1}$, $Y = A_i$, $Z = Q_i = q(A_i)$, and $W = A_{i-1}$. Because $Z = q(Y)$ is a deterministic function of Y, $H(Z \mid Y, W) = H(Z \mid X, Y, W) = 0$, so $I(X; Z \mid Y, W) = 0$. Applying the chain rule for conditional mutual information in both orders gives

$$I(X; Y, Z \mid W) = I(X; Y \mid W) + \underbrace{I(X; Z \mid Y, W)}_{=0} = I(X; Z \mid W) + I(X; Y \mid Z, W).$$

This establishes (8). Since $I(X; Y \mid Z, W) \ge 0$, (7) follows immediately, and Theorem 5.3 identifies the upper bound with d_{2b}. □

8 Non-equilibrium Markov dynamics on the codon torus

Let $P = (P_{xy})_{x,y\in\mathcal{C}}$ be a time-homogeneous Markov transition matrix on the codon torus $\mathcal{C} = \mathbb{T}_4^3$, and let $\pi = (\pi_x)_{x\in\mathcal{C}}$ be a stationary distribution ($\pi P = \pi$). Define the forward and time-reversed stationary bigram flux matrices on $\mathcal{C} \times \mathcal{C}$ by

$$J_f(x,y) := \pi_x P_{xy}, \qquad J_r(x,y) := \pi_y P_{yx} = J_f(y,x).$$

Following Schnakenberg [12] and Seifert [13], the steady-state entropy production rate per step is

$$\sigma := \sum_{x,y\in\mathcal{C}} \pi_x P_{xy} \log \frac{P_{xy}}{P_{yx}},$$

with the extended-real convention $0\log(0/0) = 0$ and $a\log(a/0) = +\infty$ for $a > 0$. Thus $\sigma \in [0, +\infty]$.

Theorem 8.1 (Entropy production, detailed balance, and cycle affinities on $\mathbb{T}_4^3$)**.** *For any finite Markov chain on the codon torus $\mathcal{C} = \mathbb{T}_4^3$ with stationary distribution π:*

(i) The entropy production rate equals the Kullback–Leibler divergence between the forward and reverse stationary flux distributions:

$$\sigma = D_{\mathrm{KL}}(J_f \,\|\, J_r) = \frac{1}{2}\sum_{x,y\in\mathcal{C}} \big(\pi_x P_{xy} - \pi_y P_{yx}\big) \log \frac{\pi_x P_{xy}}{\pi_y P_{yx}} \ge 0.$$

(ii) By Pinsker's inequality, σ is bounded below by the L_1 and Frobenius norms of the probability flux asymmetry $J_f - J_r$:

$$\sigma \geq \frac{1}{2}\|J_f - J_r\|_1^2 \geq \frac{1}{2}\|J_f - J_r\|_F^2.$$

(iii) If $\pi_x > 0$ for all $x \in \mathcal{C}$, the following three conditions are equivalent:

(a) $\sigma = 0$;

(b) Detailed balance holds: $\pi_x P_{xy} = \pi_y P_{yx}$ for all $x, y \in \mathcal{C}$;

(c) Kolmogorov's cycle condition holds on the codon torus: for every closed cycle $\gamma = (c_1, c_2, \ldots, c_m, c_1)$ in $\mathbb{T}_4^3$, the cycle affinity vanishes:

$$\mathcal{A}(\gamma) := \log \frac{P_{c_1c_2}P_{c_2c_3}\cdots P_{c_mc_1}}{P_{c_1c_m}\cdots P_{c_3c_2}P_{c_2c_1}} = 0.$$

Proof. (i) Since $\sum_{x,y} J_f(x,y) = \sum_x \pi_x \sum_y P_{xy} = 1$ and $\sum_{x,y} J_r(x,y) = 1$, both J_f and J_r are probability measures on $\mathcal{C} \times \mathcal{C}$. Using stationarity $\sum_x \pi_x P_{xy} = \pi_y$, the stationary boundary terms cancel:

$$\sum_{x,y} \pi_x P_{xy} \log \frac{\pi_x}{\pi_y} = \sum_x \pi_x \log \pi_x - \sum_y \pi_y \log \pi_y = 0.$$

Hence $D_{\mathrm{KL}}(J_f \parallel J_r) = \sum_{x,y} \pi_x P_{xy} \log(P_{xy}/P_{yx}) = \sigma \geq 0$, and symmetrizing over $(x,y) \leftrightarrow (y,x)$ gives the second expression in (i).

(ii) Pinsker's inequality for probability measures on $\mathcal{C}\times\mathcal{C}$ states that $D_{\mathrm{KL}}(J_f \parallel J_r) \geq \frac{1}{2}\|J_f - J_r\|_1^2$, and $\|v\|_1 \geq \|v\|_2$ for any vector gives $\|J_f - J_r\|_1^2 \geq \|J_f - J_r\|_F^2$.

(iii) By (i), $\sigma = 0$ if and only if $J_f = J_r$, which is detailed balance (b). If (b) holds, multiplying $\frac{P_{c_kc_{k+1}}}{P_{c_{k+1}c_k}} = \frac{\pi_{c_{k+1}}}{\pi_{c_k}}$ around any closed cycle γ telescopes to 1, giving $\mathcal{A}(\gamma) = 0$ in (c). Conversely, if (c) holds on every closed cycle, fixing a reference state c_0 on each connected component and setting $\pi_x \propto \pi_{c_0} \prod_{e \in \mathrm{path}(c_0 \to x)} (P_{e_+}/P_{e_-})$ is path-independent and satisfies (b) [12]. □

Proposition 8.2 (Support-symmetry criterion and asymptotic ε-scaling of regularized entropy production)**.** *Let P be a Markov transition matrix on $\mathcal{C}$ with stationary distribution $\pi > 0$, and define the bidirectional edge set $E_{\mathrm{bidir}} := \{(x,y) : P_{xy} > 0 \text{ and } P_{yx} > 0\}$, the unidirectional edge set $E_{\mathrm{unidir}} := \{(x,y) : P_{xy} > 0 \text{ and } P_{yx} = 0\}$, and the total unidirectional probability flux*

$$F_{\mathrm{unidir}} := \sum_{(x,y) \in E_{\mathrm{unidir}}} \pi_x P_{xy}.$$

Then:

(i) The exact entropy production σ_{exact} is finite if and only if $E_{\mathrm{unidir}} = \varnothing$ ($F_{\mathrm{unidir}} = 0$), in which case $\sigma_{\mathrm{exact}} = \sigma_{\mathrm{bidir}} := \sum_{(x,y) \in E_{\mathrm{bidir}}} \pi_x P_{xy} \log(P_{xy}/P_{yx}) < +\infty$.

(ii) Under Dirichlet transition smoothing $P_{xy}^{(\varepsilon)} = (N_{xy} + \varepsilon)/(N_x + 64\varepsilon)$ as $\varepsilon \downarrow 0$, the regularized entropy production $\sigma_{\mathrm{reg}}(\varepsilon)$ obeys the exact asymptotic expansion

$$\sigma_{\mathrm{reg}}(\varepsilon) = \sigma_{\mathrm{bidir}} + F_{\mathrm{unidir}} \log \frac{1}{\varepsilon} + O(1) \qquad (\varepsilon \downarrow 0).$$

Proof. Under the extended-real convention $a \log(a/0) = +\infty$ for $a > 0$, a summand $\pi_x P_{xy} \log(P_{xy}/P_{yx})$ diverges if and only if $(x,y) \in E_{\mathrm{unidir}}$, proving (i). For (ii), as $\varepsilon \downarrow 0$, $P_{xy}^{(\varepsilon)} \to P_{xy}$ on E_{bidir}, while for $(x,y) \in E_{\mathrm{unidir}}$, $P_{yx}^{(\varepsilon)} = \varepsilon/(N_y + 64\varepsilon) = \Theta(\varepsilon)$, so $\pi_x^{(\varepsilon)} P_{xy}^{(\varepsilon)} \log(P_{xy}^{(\varepsilon)}/P_{yx}^{(\varepsilon)}) = \pi_x P_{xy} \log(1/\varepsilon) + O(1)$. Summing over $(x,y) \in E_{\mathrm{unidir}}$ yields the stated expansion. □

8.1 Dynamic mirror balance versus static Chargaff parity [3, 4, 5, 17]

Let $\rho : \mathbb{T}_4^3 \to \mathbb{T}_4^3$ be the reverse-complement involution. If a double-stranded DNA segment is statistically indistinguishable from its reverse-complement strand at the level of directed codon bigrams, the forward flux of $x \to y$ must equal the forward flux of the reverse-complemented transition $\rho(y) \to \rho(x)$.

Definition 8.3 (Dynamic mirror current and balance)**.** The *mirror probability current* on $\mathbb{T}_4^3 \times \mathbb{T}_4^3$ is

$$J_{xy}^{\mathrm{mir}} := \pi_x P_{xy} - \pi_{\rho(y)} P_{\rho(y),\rho(x)}.$$

The Markov chain satisfies *dynamic mirror balance* if $J_{xy}^{\mathrm{mir}} = 0$ for all $x, y \in \mathbb{T}_4^3$.

Proposition 8.4 (Dynamic mirror balance implies static mirror symmetry)**.** *If a stationary Markov chain on $\mathbb{T}_4^3$ satisfies dynamic mirror balance ($J_{xy}^{\mathrm{mir}} = 0$ for all $x, y \in \mathbb{T}_4^3$), then its stationary distribution π belongs to the 32-dimensional mirror-even subspace V_+ of Corollary 2.6:*

$$\pi_x = \pi_{\rho(x)} \qquad \forall\, x \in \mathbb{T}_4^3, \qquad \textit{so} \qquad \mathrm{Mirror}_{L_1}(\pi) = 0.$$

The converse does not hold: a chain can have a mirror-symmetric stationary distribution $\pi \in V_+$ while possessing non-zero directed mirror currents $J^{\mathrm{mir}} \neq 0$ and positive entropy production $\sigma > 0$.

Proof. Summing $\pi_x P_{xy} = \pi_{\rho(y)} P_{\rho(y),\rho(x)}$ over $y \in \mathbb{T}_4^3$ and substituting $z = \rho(y)$ (since ρ is a bijection) gives

$$\pi_x = \sum_{y \in \mathbb{T}_4^3} \pi_x P_{xy} = \sum_{z \in \mathbb{T}_4^3} \pi_z P_{z,\rho(x)} = (\pi P)_{\rho(x)} = \pi_{\rho(x)}.$$

To see that the converse fails, pair the 32 mirror orbits of Corollary 2.6 into 16 disjoint pairs of orbits $\{M_{2k-1}, M_{2k}\}$, pick representatives $c_1 \in M_{2k-1}$ and $c_2 \in M_{2k}$, and form 16 disjoint 4-cycles $(c_1, c_2, \rho(c_1), \rho(c_2))$. Let P be the deterministic cyclic shift $c_1 \to c_2 \to \rho(c_1) \to \rho(c_2) \to c_1$ on each 4-cycle. Its stationary distribution is uniform ($\pi_x = 1/64$, so $\pi_x = \pi_{\rho(x)}$ and $\mathrm{Mirror}_{L_1}(\pi) = 0$), yet $P_{c_1 c_2} = 1$ while $P_{\rho(c_2),\rho(c_1)} = 0$, so $J_{c_1 c_2}^{\mathrm{mir}} = 1/64 \neq 0$ and $\sigma = +\infty > 0$. □

Proposition 8.5 (Duplex strand-asymmetry Kullback–Leibler divergence)**.** *Let $J_f(x, y) := \pi_x P_{xy}$ be the sense $5' \to 3'$ stationary bigram distribution and $J_\rho(x, y) := \pi_{\rho(y)} P_{\rho(y),\rho(x)}$ be the antiparallel antisense $5' \to 3'$ stationary bigram distribution on $\mathbb{T}_4^3 \times \mathbb{T}_4^3$. Then J_ρ is a valid probability measure on $\mathbb{T}_4^3 \times \mathbb{T}_4^3$, and the duplex strand-asymmetry divergence*

$$\sigma_{\mathrm{dup}} := D_{\mathrm{KL}}(J_f \parallel J_\rho) = \sum_{x,y \in \mathbb{T}_4^3} \pi_x P_{xy} \log \frac{\pi_x P_{xy}}{\pi_{\rho(y)} P_{\rho(y),\rho(x)}} \geq \frac{1}{2} \|J^{\mathrm{mir}}\|_1^2 \geq 0$$

vanishes ($\sigma_{\mathrm{dup}} = 0$) if and only if dynamic mirror balance holds ($J_{xy}^{\mathrm{mir}} = 0$ for all $x, y \in \mathbb{T}_4^3$).

Proof. Since ρ is a bijection and $\pi P = \pi$, $\sum_{x,y} J_\rho(x, y) = \sum_y \pi_{\rho(y)} \sum_x P_{\rho(y),\rho(x)} = \sum_z \pi_z = 1$. Since $J_f - J_\rho = J^{\mathrm{mir}}$, Pinsker's inequality gives $D_{\mathrm{KL}}(J_f \parallel J_\rho) \geq \frac{1}{2}\|J^{\mathrm{mir}}\|_1^2 \geq 0$, with equality if and only if $J_f = J_\rho$. □

9 Analytic real-data audit without Monte Carlo simulations

The theorems proved above yield a completely analytic, simulation-free statistical audit on real genomic sequences. Given a collection of coding sequences partitioned into a training set and a held-out test set, let

$$N(x, a, y) := \#\{i : C_{i-1} = x,\ A_i = a,\ C_i = y\} \qquad (x \in \mathcal{C},\ a \in \mathcal{A},\ y \in F_a)$$

denote the empirical transition counts, with marginals $N(x,a) = \sum_{y \in F_a} N(x,a,y)$, $N(a,y) = \sum_{x \in \mathcal{C}} N(x,a,y)$, and $N(a) = \sum_{x \in \mathcal{C}, y \in F_a} N(x,a,y)$.

To test the null hypothesis of zero contextual synonymous program information,

$$H_0 : S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i) = 0 \iff C_{i-1} \perp C_i \mid A_i,$$

the exact maximum-likelihood expected counts under H_0 within each synonymous fiber F_a are

$$E(x,a,y) = \frac{N(x,a)\,N(a,y)}{N(a)}.$$

The stratified log-likelihood-ratio (G^2) test statistic across all channels $a \in \mathcal{A}$ is [24]:

$$G^2 := 2 \sum_{a \in \mathcal{A}} \sum_{x \in \mathcal{C}} \sum_{y \in F_a} N(x,a,y) \log \frac{N(x,a,y)}{E(x,a,y)} = 2\, N_{\text{total}}\, \widehat{S}_{\text{syn}}, \tag{9}$$

with the convention $0 \log(0/E) = 0$, where $N_{\text{total}} = \sum_{a \in \mathcal{A}} N(a)$ is the total number of codon transitions and $\widehat{S}_{\text{syn}} = \widehat{I}(C_{i-1}; C_i \mid A_i)$ is the plug-in empirical synonymous program information in nats/codon. Analogously, for the strictly pairwise-conditioned null hypothesis $H_0^{(2)} : S_{\text{syn}}^{(2)} = I(C_{i-1}; C_i \mid A_{i-1}, A_i) = 0$, the contingency table is stratified by ordered pairs $(a,b) \in \mathcal{A} \times \mathcal{A}$ with cell counts $N(a,x,b,y)$ $(x \in F_a, y \in F_b)$ and statistic $G^2_{(2)} = 2N_{\text{total}}\widehat{S}^{(2)}_{\text{syn}}$, while for the Kemeny–Snell lumpability null hypothesis $H_0^{\text{lump}} : C_{i-1} \perp A_i \mid A_{i-1}$ $(d_{2b} = 0)$, the contingency table is stratified by predecessor channel $a \in \mathcal{A}$ with cell counts $N(x,b)$ $(x \in F_a, b \in \mathcal{A})$ and statistic $G^2_{\text{lump}} = 2N_{\text{total}}\widehat{d}_{2b}$.

Proposition 9.1 (Exact degrees of freedom for the stratified G^2, $G^2_{(2)}$, and G^2_{lump} audits)**.** *Under their respective conditional independence null hypotheses, G^2, $G^2_{(2)}$, and G^2_{lump} are sums of independent contingency-table likelihood-ratio statistics with exact degrees of freedom:*

(i) ***Full*** $64 \to 21$ ***torus convention for*** G^2 *(all 64 predecessor codons $x \in \mathcal{C}$ and all 21 channels $a \in \mathcal{A}$ active):*

$$\text{df}_{64\to21}(G^2) = \sum_{a \in \mathcal{A}} (64-1)(k_a - 1) = 63 \sum_{a \in \mathcal{A}} (k_a - 1) = 63 \times 43 = 2709.$$

(ii) ***Internal sense-codon convention*** $61 \to 20$ ***for*** G^2 *(start and stop codons trimmed, so C_{i-1} and C_i range over the 61 sense codons and 20 amino acids):*

$$\text{df}_{61\to20}(G^2) = \sum_{a \in \mathcal{A}_{\text{sense}}} (61-1)(k_a - 1) = 60 \times 41 = 2460.$$

(iii) ***Pairwise-conditioned synonymous test*** $G^2_{(2)}$ *(testing $C_{i-1} \perp C_i \mid A_{i-1}, A_i$ across all ordered channel pairs (a,b)):*

$$\text{df}_{64\to21}\big(G^2_{(2)}\big) = \sum_{a,b \in \mathcal{A}} (k_a - 1)(k_b - 1) = \left(\sum_{a \in \mathcal{A}} (k_a - 1) \right)^2 = 43^2 = 1849,$$

and in the internal sense-codon convention $61 \to 20$*:*

$$\text{df}_{61\to20}\big(G^2_{(2)}\big) = \left(\sum_{a \in \mathcal{A}_{\text{sense}}} (k_a - 1) \right)^2 = 41^2 = 1681.$$

(iv) ***Kemeny–Snell Markov lumpability test*** G^2_{lump} *(testing $C_{i-1} \perp A_i \mid A_{i-1}$, i.e. $d_{2b} = 0$, across all predecessor fibers F_a):*

$$\text{df}_{64\to21}\big(G^2_{\text{lump}}\big) = \sum_{a \in \mathcal{A}} (k_a - 1)(21 - 1) = 43 \times 20 = 860,$$

and in the internal sense-codon convention $61 \to 20$ *(with stop codons excluded from internal positions):*

$$\mathrm{df}_{61\to 20}\big(G^2_{\mathrm{lump}}\big) = \sum_{a\in\mathcal{A}_{\mathrm{sense}}} (k_a - 1)(20 - 1) = 41 \times 19 = 779.$$

(v) ***Empirical support convention*** *(when R_s rows and C_s columns are non-empty in stratum s):*

$$\mathrm{df}_{\mathrm{eff}} = \sum_{s:\, R_s>1,\, C_s>1} (R_s - 1)(C_s - 1).$$

Proof. In each stratum s with R_s active rows and C_s active columns, testing independence in an $R_s \times C_s$ contingency table has $(R_s - 1)(C_s - 1)$ degrees of freedom [24]. Summing over disjoint strata $a \in \mathcal{A}$ with $R_a = 64, C_a = k_a$ gives $63 \times 43 = 2709$; with $R_a = 61, C_a = k_a$ over $a \in \mathcal{A}_{\mathrm{sense}}$ gives $60 \times 41 = 2460$; summing $(k_a - 1)(k_b - 1)$ over ordered pairs (a, b) factors as $\big(\sum_a (k_a - 1)\big)^2$, yielding $43^2 = 1849$ and $41^2 = 1681$; and for lumpability in stratum a, the table of transitions from $x \in F_a$ (k_a rows) to next channel $b \in \mathcal{A}$ ($|\mathcal{A}|$ columns) has $(k_a - 1)(|\mathcal{A}| - 1)$ degrees of freedom, whose sum over a is $43 \times 20 = 860$ in the $64 \to 21$ convention and $41 \times 19 = 779$ in the $61 \to 20$ sense convention. □

Remark 9.2 (Sparse strata and exact alternatives)**.** The asymptotic χ^2_{df} calibration is appropriate when expected cell counts are sufficiently large. In sparse real-data strata, one may instead use exact conditional likelihood-ratio tests, analytic Dirichlet-multinomial posterior predictive checks, or category collapsing based on pre-registered support thresholds ($\mathrm{df}_{\mathrm{eff}}$). These alternatives preserve the simulation-free analytic logic and do not alter the exact degrees-of-freedom derivations under full-support conventions.

Remark 9.3 (Clustered sequence data and analytic calibration)**.** The asymptotic χ^2_{df} calibration of G^2 and $G^2_{(2)}$ is direct under independent multinomial sampling within strata. In real genomic data, codon transitions within the same gene are clustered (sharing gene-level GC content, expression level, and evolutionary history). Therefore, for conservative inference one should either: (i) treat the pooled G^2 and $\widehat{S}_{\mathrm{syn}}$ as likelihood-ratio effect-size diagnostics rather than raw unadjusted p-values; (ii) compute gene-level log-likelihood-ratio contributions on held-out genes after family-aware deduplication and combine independent genes analytically (e.g. via a nonparametric sign/Wilcoxon test or Fisher's method across independent loci); or (iii) use a cluster-robust sandwich variance estimator for $\widehat{S}_{\mathrm{syn}}$ with genes (or paralog families) as independent clusters. These procedures preserve a 100% analytic, simulation-free workflow while preventing anti-conservative significance inflation from within-gene dependence.

Alongside the contingency tests, out-of-sample generalization is audited analytically on held-out test genes using Dirichlet-smoothed training distributions ($\alpha > 0$):

$$\Delta\mathrm{LL}_{\mathrm{syn}} := \frac{1}{N_{\mathrm{test}}} \sum_{i\in\mathrm{test}} \log \frac{\widehat{P}_{\mathrm{train}}(C_i \mid C_{i-1}, A_i)}{\widehat{P}_{\mathrm{train}}(C_i \mid A_i)} = d_2 - d_{2b},$$

which estimates S_{syn} out-of-sample without a single random number generator.

9.1 Real-data sources for the simulation-free audit

The analytic protocol can be instantiated without synthetic surrogate generation using public biological databases:

- **Genomic and transcriptomic coding sequences:** NCBI RefSeq CDS (`*_cds_from_genomic.fna.gz`), Telomere-to-Telomere human reference `T2T-CHM13v2.0`, GENCODE, Ensembl, and MANE Select transcripts for held-out coding sequences;
- **Gene structure and splicing compartments:** NCBI RefSeqGene (`NG_*`) records for four-state exon/intron compartment and splice-junction phase audits;

- **Protein structure and physicochemical coarsenings:** UniProt, Protein Data Bank (PDB), AlphaFold Protein Structure Database, DisProt (intrinsically disordered regions), Pfam/InterPro, and PhosphoSitePlus for deterministic coarsenings $q : \mathcal{A} \to \mathcal{Q}$;
- **Translation kinetics and tRNA pools:** GtRNAdb (genomic tRNA gene copy numbers) and ribosome-profiling archives (GWIPS-viz and processed Ribo-seq dwell-time matrices from GEO/SRA);
- **Somatic and germline variation:** COSMIC, ClinVar, gnomAD, and dbSNP for auditing single-codon synonymous and missense variants;
- **Host–virus specificity:** NCBI Virus and ICTV host metadata for cross-scoring viral coding sequences against host Markov transition kernels.

In all cases, primary evidence is obtained by evaluating real held-out biological sequences under analytic log-likelihood, contingency-table, and Markov-flow operators, rather than Monte Carlo surrogate simulations.

10 Relation to classical codon-usage measures and compartment audits

Classical codon-usage indices—such as the Codon Adaptation Index (CAI) [9], the tRNA Adaptation Index (tAI), and the Effective Number of Codons (ENC)—summarize zero-order marginal codon frequencies $P(C)$ or fiber-level marginal distributions $P(C \mid A = a) \in \Delta_{\text{syn}}$. In the terminology of [16, 17, 23] and the present work, these indices characterize the static *form* of the genetic-code fibration.

By contrast, the quantities isolated in Theorems 5.3 and 5.4,

$$d_{2b} = I(C_{i-1}; A_i \mid A_{i-1}), \qquad S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i), \qquad S^{(2)}_{\text{syn}} = I(C_{i-1}; C_i \mid A_{i-1}, A_i),$$

measure first-order sequential dependencies that survive after the amino-acid channel is conditioned out. Theorem 6.1 proves that deterministic Max-CAI optimization maximizes zero-order marginal adaptation at the cost of annihilating all three conditional sequential layers.

The same analytical decomposition extends directly to two key biological applications:

1. **Ribosomal dwell-time association:** If empirical ribosome-profiling dwell times τ_i at position i satisfy $I(C_{i-1}; \tau_i \mid A_i, \text{local controls}) > 0$ on held-out transcripts, this provides analytic evidence that the conditional synonymous layer Δ_{syn} is coupled to translational kinetics.
2. **Exon–intron compartment contrasts:** Because Markov log-likelihoods are additive over observed triplet transitions, the quantities d_{2b}, S_{syn}, Mirror_{L_1}, and σ can be evaluated separately on concatenated coding exons, unspliced pre-mRNA spans (across all three reading frames), excised introns, and splice-junction boundaries in RefSeqGene records without generating synthetic intron or exon surrogates.

11 Empirical hourglass profiles and the genomics–proteomics duality on genomic windows

The algebraic decomposition of the codon torus proved above provides a rigorous operator-theoretic foundation for interpreting macroscopic scalar profiles computed along chromosomal windows [16, 18, 19, 20, 21, 22]. Let $w_i \in \Delta_{64}$ denote the empirical codon-frequency vector in a window indexed by i, let $\alpha_* : \Delta_{64} \to \Delta_{21}$ denote the pushforward map summing codon frequencies over each synonymous fiber F_a, and let $\alpha^* : \mathbb{R}^{21} \hookrightarrow \mathbb{R}^{64}$ denote the pullback embedding functions on $\mathcal{A}$ as fiberwise-constant functions on $\mathcal{C}$.

Given a genomic scalar functional $\varphi : \Delta_{64} \to \mathbb{R}$ (such as the conditional log-probability potential $s(c) = \log P_{\text{train}}(c)$) and its amino-acid conditional expectation $\psi(a) = \sum_{y \in F_a} P_{\text{train}}(y \mid A = a)\, s(y)$, define the genomic window profile $G(i) := \langle w_i, s \rangle$, the proteomic pushforward profile $P(i) := \langle \alpha_* w_i, \psi \rangle = \langle w_i, \alpha^* \psi \rangle$, and the antisense reverse-complement view $G^\rho(i) := \langle \rho^* w_i, s \rangle$.

Proposition 11.1 (Sense–antisense coincidence and the mirror-even "father" projection)**.** *For any window frequency vector $w_i \in \Delta_{64}$, the sense and antisense evaluations coincide for all mirror-symmetric potentials if and only if $w_i \in V_+$, or equivalently* $\mathrm{Mirror}_{L_1}(w_i) = \|P_{V_-} w_i\|_1 = 0$*. Thus the strand-invariant "father" profile is the mirror-even projection $P_{V_+} w_i \in V_+$, while the discrepancy between the two strand views is governed by the mirror-odd projection $P_{V_-} w_i \in V_-$.*

Proof. By Corollary 2.6(ii), $\mathbb{R}^{64} = V_+ \oplus V_-$ with $P_{V_\pm} = \frac{1}{2}(I \pm \rho^*)$. Hence $w_i - \rho^* w_i = 2P_{V_-} w_i$, which vanishes if and only if $w_i \in V_+$. □

Theorem 11.2 (L^2-orthogonal projection and Pythagorean variance decomposition on the genetic–code fibration)**.** *Let P be a strictly positive probability measure on $\mathcal{C} = \mathbb{T}_4^3$, and equip $\mathbb{R}^{\mathcal{C}} \cong \mathbb{R}^{64}$ with the weighted inner product $\langle f, g \rangle_{L^2(P)} := \sum_{c \in \mathcal{C}} P(c) f(c) g(c)$. Define the conditional expectation operator*

$$(\Pi_\alpha s)(c) := \mathbb{E}_P\big[s(C) \mid \alpha(C) = \alpha(c)\big] = (\alpha^* \psi)(c), \qquad \psi(a) := \sum_{y \in F_a} P(y \mid A = a)\, s(y).$$

Then Π_α is the unique $L^2(P)$-orthogonal projection onto the 21-dimensional pullback subspace $\mathrm{Im}(\alpha^*) \subset L^2(\mathcal{C}, P)$, its orthogonal complement $\ker(\Pi_\alpha)$ (the intra-fiber synonymous potential space) has exact real dimension $64 - 21 = 43$ (or $61 - 20 = 41$ on sense codons), and for every potential $s \in L^2(\mathcal{C}, P)$ the total variance decomposes orthogonally as

$$\mathrm{Var}_P\big(s(C)\big) = \underbrace{\mathrm{Var}_P\big(\psi(A)\big)}_{\textit{Amino-acid form variance}} + \underbrace{\mathbb{E}_P\big[\mathrm{Var}_P(s(C) \mid A)\big]}_{\textit{Synonymous fiber variance}}.$$

Proof. Clearly $\Pi_\alpha^2 = \Pi_\alpha$ and $\mathrm{Im}(\Pi_\alpha) = \mathrm{Im}(\alpha^*)$, which has dimension $|\mathcal{A}| = 21$ because α is surjective (so α^* is injective). By the rank-nullity theorem, $\dim \ker(\Pi_\alpha) = 64 - 21 = 43$. For any $f, g \in L^2(\mathcal{C}, P)$, the tower property of conditional expectation gives

$$\langle \Pi_\alpha f, g \rangle_{L^2(P)} = \mathbb{E}_P\big[\mathbb{E}_P[f(C) \mid A]\, g(C)\big] = \mathbb{E}_P\big[\mathbb{E}_P[f(C) \mid A]\, \mathbb{E}_P[g(C) \mid A]\big] = \langle f, \Pi_\alpha g \rangle_{L^2(P)},$$

so Π_α is self-adjoint and idempotent, hence an orthogonal projector. Applying the Pythagorean theorem to $s - \mathbb{E}_P[s] = (\Pi_\alpha s - \mathbb{E}_P[s]) + (s - \Pi_\alpha s)$ yields the law of total variance. □

Lemma 11.3 (Window marginal aggregation erases the sequential bigram program)**.** *Any profile functional $G(i) = \varphi(w_i)$ that depends solely on the zero-order marginal codon-frequency vector w_i of a window is invariant under permutations of codon order within the window and therefore does not determine the conditional synonymous program information $S_{\mathrm{syn}} = I(C_{i-1}; C_i \mid A_i)$ or $S_{\mathrm{syn}}^{(2)} = I(C_{i-1}; C_i \mid A_{i-1}, A_i)$.*

Proof. Let $F_a = \{x, y\}$ be a 2-codon synonymous fiber and consider two windows $W_1 = (x, y, x, y, \ldots, x, y)$ and $W_2 = (x, x, \ldots, x, y, y, \ldots, y)$ of equal length $2m$. Both windows have the exact same marginal codon vector $w(x) = w(y) = \frac{1}{2}$ (so $G(W_1) = G(W_2)$ and $P(W_1) = P(W_2)$), yet W_1 alternates deterministically ($P(y \mid x, a) = P(x \mid y, a) = 1$, maximizing $S_{\mathrm{syn}} = \log 2$), whereas W_2 has only a single $x \to y$ transition across $2m - 1$ steps. Thus window marginals cannot factor S_{syn}. □

Remark 11.4 (Resolution separation: macroscopic form versus microscopic program)**.** Lemma 11.3 resolves why a near-unity cumulative profile concordance between $G(i)$ and $P(i)$ along a 100-kb chromosomal span (confirming that macroscopic compositional *form* is captured by the amino-acid pushforward α_*) coexists with a strictly positive sequential bigram residual $S_{\mathrm{syn}} > 0$ and $d_{2b} > 0$: the two diagnostics operate on orthogonal levels of the Markov hierarchy (zero-order window marginals versus first-order conditional transitions).

11.1 Guarded origin hypothesis: analytical test for double-strand relic coupling

The two-strand channel coupling is an *empirical double-strand fact* rather than a hypothesis: for every real duplex genome, the sense channel $A(c) = \alpha(c)$ and the antisense channel $(A \circ \rho)(c) = \alpha(\rho(c))$ are evaluated on the same 21×21 contingency table, and the resulting mutual information

$$I(A;\, A \circ \rho) = \sum_{a,b \in \mathcal{A}} P\big(\alpha(C) = a,\, \alpha(\rho(C)) = b\big) \log \frac{P\big(\alpha(C) = a,\, \alpha(\rho(C)) = b\big)}{P(\alpha(C) = a)\, P(\alpha(\rho(C)) = b)} > 0$$

is strictly positive across all nine audited taxa (Section 12), tested analytically with exact full-support degrees of freedom $(21-1)^2 = 400$ (or $(20-1)^2 = 361$ on the 20 sense amino-acid channels). This positivity is a structural consequence of the double-stranded fibration $\alpha_{\mathrm{dup}} = (\alpha, \alpha \circ \rho)$ proved in Theorem 4.3 and requires no abiogenetic assumption. The epistemic guard is therefore confined *solely* to the evolutionary *origin* interpretation of that coupling—namely, the phylogenetic hypothesis [16, 17, 22] that the 32 mirror codon pairs originated from an ancestral double-stranded coding duplex that subsequently unfolded into single-stranded primers, which we keep strictly separated from the proved algebraic theorems on $\mathbb{T}^3_4$.

12 Empirical validation across nine RefSeq genomes and the human 17q22 locus

We executed the complete simulation-free analytical audit (`codon_torus_real_audit_v3.py`, version `v3.2`) on all deduplicated coding sequences (`-max-records 0`, strict true-frame parsing, start/stop codons trimmed, internal-stop pseudogenes filtered, one-gene-one-vote longest-isoform retention, and deterministic `SHA-256` 80/20 `family_root` gene splits) across nine reference genomes from NCBI RefSeq spanning Eukarya, Bacteria, and Archaea: *Homo sapiens* (`T2T-CHM13v2.0`, `GCF_009914755.1`), *Mus musculus* (`GRCm39`, `GCF_000001635.27`), *Drosophila melanogaster* (`Release_6_plus_ISO1_MT`, `GCF_000001215.4`), *Arabidopsis thaliana* (`TAIR10.1`, `GCF_000001735.4`), *Saccharomyces cerevisiae* (`R64`, `GCF_000146045.2`), *Escherichia coli* K-12 (`ASM584v2`, `GCF_000005845.2`), *Bacillus subtilis* 168 (`ASM904v1`, `GCF_000009045.1`), *Methanocaldococcus jannaschii* (`ASM774v1`, `GCF_000007745.1`), and *Halobacterium salinarum* (`ASM1558v1`, `GCF_000015585.1`), totaling 9,834,024 held-out test codon transitions.

Species	Tab	Test genes	Test trans	$I(C;C)$	$I(A;A)$	d_{2b}	S_{syn}	$I(A_1;C_2\|A_2)$	$S^{(2)}_{\mathrm{syn}}$	Residual	df($G^2/G^2_{(2)}$/lump)	Gene t-stat (pos%)
`Human_T2T`	1	3,999	2,342,893	+0.0842	+0.0099	+0.0483	+0.0260	+0.0038	+0.0222	4.2×10^{-17}	2460/1681/779	+52.58 (83.0%)
`M_musculus`	1	4,509	2,411,478	+0.0770	+0.0091	+0.0543	+0.0136	+0.0032	+0.0104	2.2×10^{-16}	2460/1681/779	+49.46 (84.5%)
`D_melanogaster`	1	2,698	1,493,418	+0.0632	+0.0125	+0.0343	+0.0164	+0.0061	+0.0103	1.9×10^{-16}	2460/1681/779	+54.63 (87.7%)
`A_thaliana`	1	5,426	2,186,312	+0.0458	+0.0082	+0.0253	+0.0123	+0.0040	+0.0083	2.8×10^{-17}	2460/1681/779	+64.57 (85.1%)
`S_cerevisiae`	1	1,171	563,029	+0.0254	+0.0065	+0.0111	+0.0078	+0.0026	+0.0053	3.5×10^{-17}	2460/1681/779	+17.43 (73.6%)
`E_coli`	11	875	272,548	+0.0943	+0.0103	+0.0475	+0.0366	+0.0162	+0.0204	2.8×10^{-17}	2460/1680/779	+33.85 (90.1%)
`B_subtilis`	11	897	251,034	+0.0811	+0.0102	+0.0439	+0.0269	+0.0090	+0.0179	1.9×10^{-16}	2460/1681/779	+27.98 (86.7%)
`M_jannaschii`	11	439	147,216	+0.1152	+0.0101	+0.0660	+0.0392	+0.0137	+0.0255	3.1×10^{-16}	2456/1676/779	+30.51 (91.8%)
`H_salinarum`	11	480	166,096	+0.0721	+0.0085	+0.0387	+0.0249	+0.0125	+0.0125	6.9×10^{-17}	2455/1671/779	+17.09 (85.4%)

Table 2: Exact tripartite and quadripartite mutual-information decompositions (in nats/codon), contingency-table degrees of freedom (df for $G^2(S_{\mathrm{syn}})$, $G^2_{(2)}(S^{(2)}_{\mathrm{syn}})$, and $G^2_{\mathrm{lump}}(d_{2b})$; all $p < 10^{-300}$), and cluster-robust held-out gene-level t-statistics across nine RefSeq genomes (`v3.2`, zero simulations).

Tables 2 and 3 and Figures 1–2 establish six empirical conclusions:

1. **Machine-precision chain-rule identities and exact degrees of freedom:** Across all nine species, the tripartite and quadripartite decompositions hold with numerical residuals between 2.8×10^{-17} and 3.1×10^{-16} nats/codon (Table 2, Fig. 1a,b). With internal-stop pseudogenes removed, the empirical degrees of freedom never exceed the pre-registered full-support sense-codon bounds of Proposition 9.1: $\mathrm{df}(G^2) = 2460$ in 7/9 species (dropping slightly to 2456 and 2455 in the two sparsest archaeal genomes due to unobserved rare codon pairs, Remark 9.2), $\mathrm{df}(G^2_{(2)}) = 1681$ in 6/9 species, and $\mathrm{df}(G^2_{\mathrm{lump}}) = 779$ in all 9/9 species (Fig. 2d).

Species	d_{2b}	I_{Hydro}	I_{Chg}	I_{Vol}	I_{Arom}	Unidir edges	σ_{exact}	σ_{reg}	$\|J^{\text{mir}}\|_1$	$\text{Mirror}_{L_1}(\pi)$	Max-CAI Int (S_{syn}, d_{2b})	Cross d_{2b} WT $\to$ Max-CAI ($< 0\%$)
Human_T2T	+0.0483	+0.0015	+0.0159	+0.0109	+0.0029	0	+0.1276	+0.1276	0.6169	0.2002	(0.0000, 0.0000)	+0.0550 $\to$ −0.0581 (96.0%)
M_musculus	+0.0543	+0.0014	+0.0172	+0.0135	+0.0033	0	+0.1441	+0.1441	0.5851	0.1904	(0.0000, 0.0000)	+0.0530 $\to$ −0.0663 (98.7%)
D_melanogaster	+0.0343	+0.0034	+0.0036	+0.0038	+0.0025	0	+0.0860	+0.0860	0.7369	0.2338	(0.0000, 0.0000)	+0.0395 $\to$ −0.0390 (93.0%)
A_thaliana	+0.0253	+0.0022	+0.0061	+0.0070	+0.0009	0	+0.0634	+0.0634	0.6763	0.2418	(0.0000, 0.0000)	+0.0261 $\to$ −0.0254 (94.3%)
S_cerevisiae	+0.0111	+0.0017	+0.0028	+0.0017	+0.0007	0	+0.0363	+0.0363	0.8078	0.2709	(0.0000, 0.0000)	+0.0098 $\to$ −0.0078 (77.0%)
E_coli	+0.0475	+0.0101	+0.0065	+0.0092	+0.0022	17	$+\infty$	+0.1779	0.7997	0.2562	(0.0000, 0.0000)	+0.0395 $\to$ −0.0298 (94.0%)
B_subtilis	+0.0439	+0.0079	+0.0085	+0.0086	+0.0014	7	$+\infty$	+0.1608	0.9021	0.2995	(0.0000, 0.0000)	+0.0366 $\to$ −0.0292 (91.3%)
M_jannaschii	+0.0660	+0.0129	+0.0073	+0.0164	+0.0065	55	$+\infty$	+0.3025	1.0022	0.3280	(0.0000, 0.0000)	+0.0582 $\to$ −0.0418 (97.0%)
H_salinarum	+0.0387	+0.0111	+0.0032	+0.0023	+0.0040	146	$+\infty$	+0.1914	0.8461	0.2823	(0.0000, 0.0000)	+0.0302 $\to$ −0.0217 (93.3%)

Table 3: Chemical data-processing inequality audit ($0 \le I(C_{i-1}; Q_i \mid A_{i-1}) \le d_{2b}$ satisfied in 36/36 tests), exact and ε-regularized Markov entropy production (σ_{exact}, σ_{reg}), dynamic and static mirror asymmetries, and internal vs. cross-scored Max-CAI collapse across nine RefSeq genomes (`v3.2`).

2. **Rejection of $S_{\text{syn}} = 0$, $S^{(2)}_{\text{syn}} = 0$, and Kemeny–Snell lumpability ($d_{2b} = 0$):** Every species exhibits strictly positive contextual synonymous program information $S_{\text{syn}} \in [+0.0078, +0.0392]$, strictly positive pairwise-conditioned synonymous information $S^{(2)}_{\text{syn}} \in [+0.0053, +0.0255]$, and strictly positive codon-to-next-amino-acid context $d_{2b} \in [+0.0111, +0.0660]$ (G^2, $G^2_{(2)}$, and G^2_{lump} all yield $p < 10^{-300}$). Out-of-sample gene-level held-out evaluation (Remark 9.3, Fig. 2b) confirms positive mean $\Delta\text{LL}_{\text{syn}}$ in 9/9 species (73.6%–91.8% of individual test genes positive; one-sample t-statistics from +17.09 to +64.57).
3. **Dual verification of the Max-CAI collapse and pan-domain cross-scoring crash:** Recoding test genes via the deterministic Max-CAI map $C_i = \phi(A_i)$ collapses internal S_{syn}, $S^{(2)}_{\text{syn}}$, d_{2b}, and $I(A_{i-1}; C_i \mid A_i)$ identically to +0.000000 in all nine species (Theorem 6.1), while scoring those Max-CAI sequences under the wild-type transition model P_{WT} drives d_{2b} from positive wild-type means (+0.0098 to +0.0582) to strictly negative means (−0.0078 to −0.0663, with 77.0%–98.7% of individual genes negative; Proposition 6.2 and Fig. 1c).
4. **Chemical data-processing hierarchy:** Across all 9 species and all 4 physicochemical coarsenings (hydrophobicity, charge, volume, aromaticity; 36 tests), $0 < I(C_{i-1}; Q_i \mid A_{i-1}) < d_{2b}$ and the exact residual identity $I(C_{i-1}; Q_i \mid A_{i-1}) + I(C_{i-1}; A_i \mid Q_i, A_{i-1}) = d_{2b}$ hold to machine precision (Theorem 7.1, Table 3). Vertebrates prioritize electrostatic charge and side-chain volume prediction ($I_{\text{Chg}} = +0.0159/+0.0172$ in human/mouse vs. $I_{\text{Hydro}} = +0.0015/+0.0014$), whereas prokaryotes and extremophiles exhibit strong hydropathy context ($I_{\text{Hydro}} = +0.0101$ in *E. coli*, +0.0129 in *M. jannaschii*, and +0.0111 in *H. salinarum*).
5. **Two regimes of non-equilibrium entropy production and mirror currents:** All nine species violate detailed balance ($\sigma_{\text{reg}} \in [+0.0363, +0.3025]$ nats/codon) and exhibit strong dynamic mirror currents ($\|J^{\text{mir}}\|_1 \in [0.5851, 1.0022]$) and static mirror asymmetry ($\text{Mirror}_{L_1}(\pi) \in [0.1904, 0.3280]$; Theorem 8.1, Proposition 8.2, and Fig. 2a). In the five dense eukaryotic proteomes (0.56–2.41×10^6 transitions), all 61×61 sense transitions are bidirectional (unidir_edges $= 0$, so $\sigma_{\text{exact}} = \sigma_{\text{reg}} < +\infty$), whereas the four compact prokaryotic/archaeal proteomes retain 7–146 strictly unidirectional sense transitions on the raw support ($\sigma_{\text{exact}} = +\infty$).
6. **Human 17q22 100-kb hourglass anchor and relic coupling:** On the cached human chromosome 17 window `NC_000017.11:55154472-55254472` (100,001 nt, 33,333 triplets; Fig. 2c), the cumulative genomic and proteomic Master-Code distance profiles from the empirical isthmus (codon 6,805) achieve a macroscopic concordance of 99.998% (Link = 100.00%, local sliding-window $R^2_{\text{prof}} = 0.1688$, Theorem 11.2), while the sequential bigram program inside the same window remains strictly positive ($S_{\text{syn}} = +0.0539$ nats/codon), directly illustrating Lemma 11.3. Finally, the sense–antisense channel table $(\alpha(c), \alpha(\rho(c)))$ exhibits $I(A; A \circ \rho) \in [+1.8510, +1.9973]$ nats across the nine species (analytic G^2 on the 21×21 sense–antisense table of Section 11.1, full-support df $= 400$; the internal-stop guard leaves the Stop row empty while Stop antisense columns remain, so the audit JSON reports an effective $\text{df}_{\text{eff}} \le 400$; $p < 10^{-300}$).

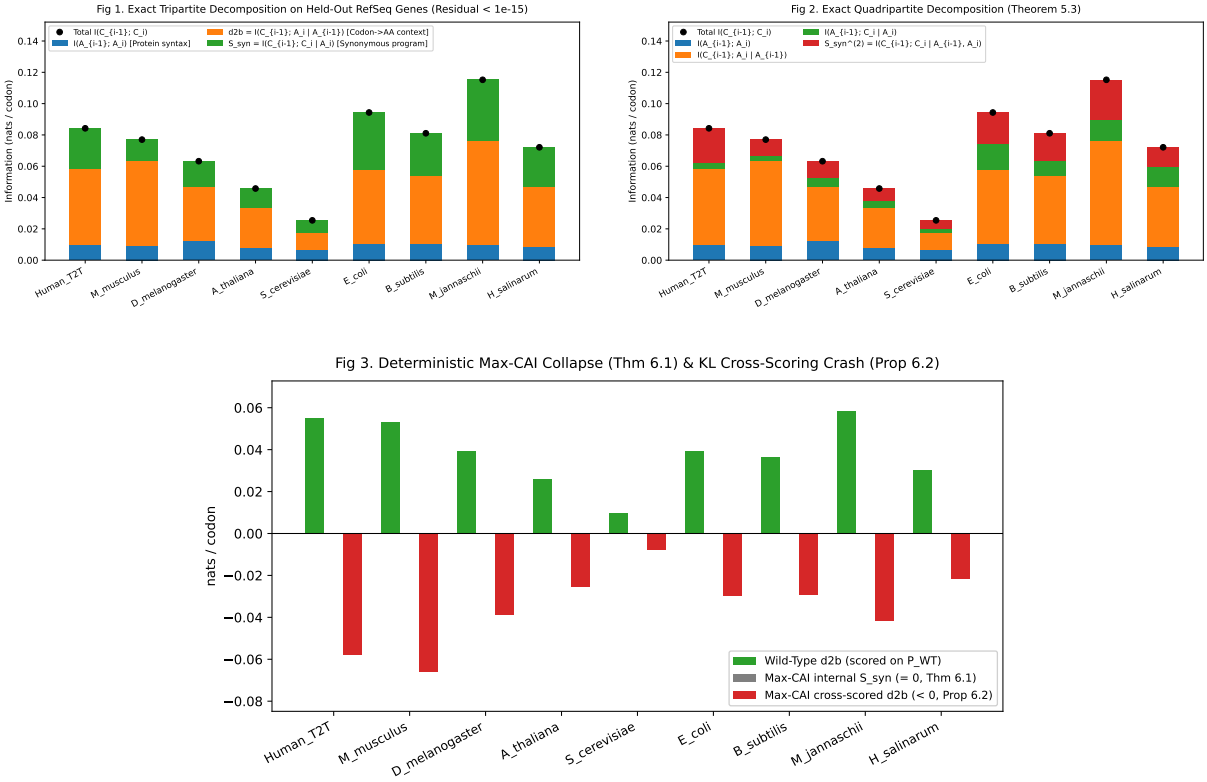


Figure 1: Simulation-free empirical verification on held-out RefSeq test genes across nine species: (top left) exact tripartite decomposition (Theorem 5.3); (top right) exact quadripartite decomposition (Theorem 5.4); (bottom) internal synonymous program collapse ($S_{\text{syn}} = 0$, Theorem 6.1) and cross-scored $d_{2b} < 0$ penalty (Proposition 6.2) under deterministic Max-CAI codon optimization.

Species	$S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i)$	$S_{\text{anti}} = I(C_{i-1}; A_i^\rho \mid A_i)$	$S_{\text{dup}} = I(C_{i-1}; C_i \mid A_i, A_i^\rho)$	Antisense share $S_{\text{anti}}/S_{\text{syn}}$	Residual $\lvert S_{\text{syn}} - (S_{\text{anti}} + S_{\text{dup}})\rvert$	$I(A; A \circ \rho)$
Human_T2T	+0.0260	+0.0215	+0.0046	**82.5**%	3.8×10^{-17}	+1.9179
M_musculus	+0.0136	+0.0110	+0.0026	**81.1**%	1.0×10^{-17}	+1.9299
D_melanogaster	+0.0164	+0.0134	+0.0030	**81.8**%	3.5×10^{-18}	+1.9065
A_thaliana	+0.0123	+0.0099	+0.0024	**80.4**%	3.5×10^{-18}	+1.9454
S_cerevisiae	+0.0078	+0.0062	+0.0016	**79.0**%	0.0×10^{0}	+1.9731
E_coli	+0.0366	+0.0304	+0.0061	**83.2**%	6.9×10^{-18}	+1.9777
B_subtilis	+0.0269	+0.0200	+0.0070	**74.1**%	2.4×10^{-17}	+1.9578
M_jannaschii	+0.0392	+0.0338	+0.0053	**86.4**%	6.9×10^{-18}	+1.9973
H_salinarum	+0.0249	+0.0221	+0.0029	**88.5**%	3.5×10^{-18}	+1.8510

Table 4: Empirical instantiation of the two-strand decomposition $S_{\text{syn}} = S_{\text{anti}} + S_{\text{dup}}$ (Theorem 5.5) and duplex channel coupling $I(A; A \circ \rho)$ (Theorem 4.3) across nine RefSeq genomes (in nats/codon). Across all nine taxa, 74.1%–88.5% of the sense-strand synonymous program is carried by the antisense amino-acid channel $A_i^\rho = \alpha(\rho(C_i))$.

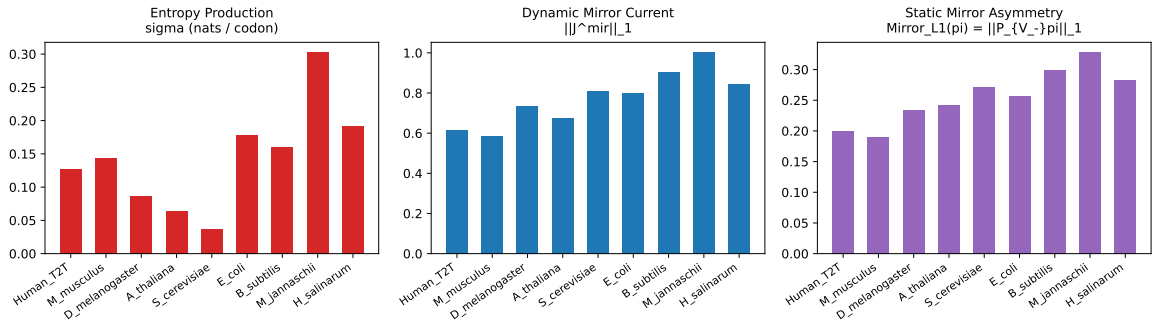
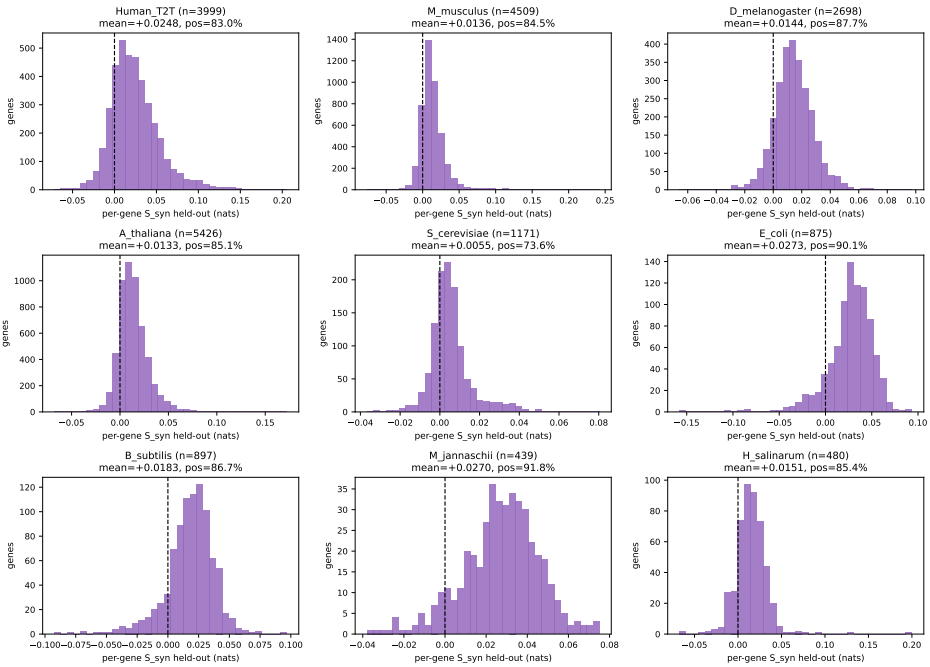
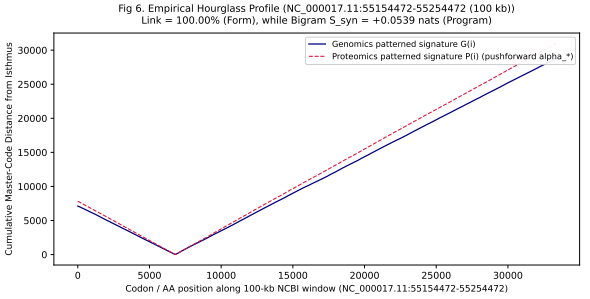


Fig 5. Cluster-Robust Gene-Level Evidence on Held-Out Test Genes





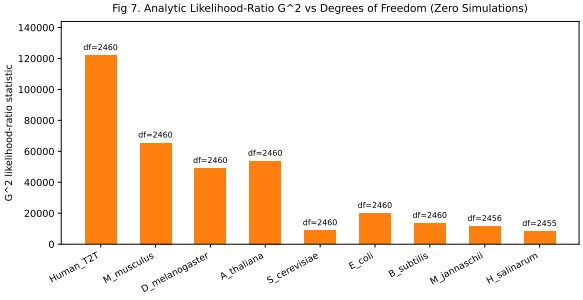


Figure 2: Non-equilibrium torus dynamics and statistical audits across nine RefSeq species: (top) steady-state entropy production σ_{reg}, dynamic mirror current $\|J^{\text{mir}}\|_1$, and static mirror asymmetry $\text{Mirror}_{L_1}(\pi)$; (middle) cluster-robust per-gene held-out $\Delta\text{LL}_{\text{syn}}$ distributions across all 9 species ($n = 439$–$5{,}426$ test genes per species); (bottom left) empirical cumulative V-profile on `NC_000017.11:55154472-55254472` (33,333 triplets); (bottom right) analytic G^2 statistics and exact degrees of freedom (df ≤ 2460; canonical $60 \cdot 41 = 2460$ in 7/9 taxa).

Two-strand instantiation (the explicit double-strand basis). All nine audited organisms are double-stranded, and the audit evaluates *both* strands through ρ rather than a single strand (Table 4). The directed two-strand flux asymmetry is strictly positive in 9/9 taxa ($\|J^{\mathrm{mir}}\|_1 \in [0.5851, 1.0022]$, so the sense and antisense strands carry different bigram currents), and the two-strand relic coupling is strictly positive in 9/9 taxa ($I(A; A \circ \rho) \in [+1.8510, +1.9973]$ nats, full-support df = 400 / sense $\mathrm{df}_{\mathrm{eff}} \leq 400$, $p < 10^{-300}$). Most remarkably, instantiating Theorem 5.5 ($S_{\mathrm{syn}} = S_{\mathrm{anti}} + S_{\mathrm{dup}}$, held-out residual $\leq 3.8 \times 10^{-17}$ nats/codon) across the nine reference genomes demonstrates that the antisense amino-acid channel program $S_{\mathrm{anti}} = I(C_{i-1}; A_i^{\rho} \mid A_i)$ accounts for **74.1**%–**88.5**% of the entire sense-strand synonymous program S_{syn} (including **82.5**% in `Human_T2T`, **83.2**% in `E_coli`, **86.4**% in `M_jannaschii`, and **88.5**% in `H_salinarum`; Table 4). Because the duplex fibration α_{dup} (Theorem 4.3) leaves only 8 binary doublets unresolved once both strand channels (A_i, A_i^{ρ}) are specified, over three-quarters of what appears on a single strand as "synonymous" sequential information is mathematically equivalent to specifying the antisense amino-acid channel on the complementary strand. At the zero-order marginal level, instantiating Theorem 4.3(iii) on held-out `Human_T2T` test genes (2,342,893 transitions) yields the exact split

$$H(C \mid A) = 1.1132 \text{ nats} = \underbrace{H(A^{\rho} \mid A)}_{0.9904 \text{ nats } (\mathbf{88.97}\%)} + \underbrace{H(C \mid A, A^{\rho})}_{0.1228 \text{ nats } (\mathbf{11.03}\%)}$$

(numerical residual $< 10^{-16}$ nats), confirming that 89.0% of single-strand marginal synonymous entropy is resolved by the antisense amino-acid channel $A^{\rho} = \alpha(\rho(C))$. Likewise, the $L^2(P)$ Pythagorean variance decomposition of Theorem 11.2 holds across all audited cohorts with numerical residuals $\leq 1.1 \times 10^{-16}$. The abiogenetic reading of this coupling remains, by design, under the Level-B guard of Section 11.1; the two-strand mathematical equivalence itself is an exact theorem.

12.1 Comparative six-frame duplex forensics, dual frame-opening, and Corollary 4.4 Viterbi compilation on BNT162b2 and Moderna mRNA-1273

To test Corollary 4.4, Theorem 6.1, Propositions 6.2–6.3, Proposition 8.5, and Theorem 11.2 across industrial synthetic mRNA constructs, we implemented the standalone zero-RNG verification suite `codon_torus_real_audit_v3_bnt162b2.py` (module `v3.5`, flag `-bnt162b2-only`). The script scores seven full-length coding sequences for the SARS-CoV-2 Spike glycoprotein against the complete `Human_T2T` reference Markov transition kernel (16,106 training genes, 9,279,815 training transitions; 3,999 held-out test genes, 2,342,893 test transitions):

(1) `WT_SARS2_Spike_NC045512`: the wild-type SARS-CoV-2 Wuhan-Hu-1 Spike coding sequence fetched live from NCBI Entrez (`NC_045512.2:21563-25384`, 3,822 nt, 1,274 codons);
(2) `BNT162b2_Full_Spike_2P`: the canonical codon-optimized Spike-2P (K986P/V987P) coding sequence of the Pfizer/BioNTech `BNT162b2` mRNA vaccine (3,825 nt, 1,275 codons including the tandem terminal `TGATGA` stop pair, verified against partial NCBI vector records `OR134577.1` and `OQ983940.1`);
(3) `mRNA1273_Full_Spike_2P`: the canonical monovalent Moderna `mRNA-1273` Spike-2P (K986P/V987P) coding sequence (3,828 nt, 1,276 codons including the triple terminal stop `TGATAATAG`, Stanford/Fire Lab reference assembly);
(4) `mRNA1273_NCBI_OR134578`: the full-length Moderna Spike expression vector ORF extracted directly from NCBI Entrez record `OR134578.1` (3,819 nt, 1,273 codons, tandem terminal stop `TGATGA`);
(5) `Human_MaxCAI_Spike_2P`: deterministic human Max-CAI recoding $C_i = \phi_{\mathrm{MaxCAI}}(A_i)$ of Spike-2P (3,822 nt, 1,274 codons, Theorem 6.1);
(6) `Human_MaxGC3_Spike_2P`: deterministic human Max-GC3 recoding (GC3 = 1.0000, selecting the most frequent G/C-ending human codon in each fiber F_a);
(7) `Human_Markov_H2b_Spike_2P`: an unconstrained 100% deterministic (zero-RNG) Viterbi dynamic-programming compilation on the synonymous fiber trellis $F_{A_0} \times \cdots \times F_{A_{m-1}} \subset$

$(\mathbb{T}_4^3)^m$ maximizing the conditional syntax excess $\sum_{i=1}^{m-1}\big[\log P_{\mathrm{WT}}(C_i \mid C_{i-1}) - \log P_{\mathrm{WT}}(A_i \mid A_{i-1}) - 0.85 \log P_{\mathrm{WT}}(C_i \mid A_i)\big]$ (Proposition 6.3);

(8) `Human_DuplexSafe_Viterbi_Spike_2P`: a constructive Corollary 4.4 ambigram-guarded Viterbi compiler (3,822 nt, 1,274 codons) that restricts the Histidine fiber to CAC (#{CAT} = 0, eliminating all antisense frame-−0 ATG = ρ(CAT) start sites) while restoring periodic antisense termination barriers from $F_{\mathrm{Leu,Stop}} \cup F_{\mathrm{Ser,Stop}} = \{\mathrm{TTA}, \mathrm{CTA}, \mathrm{TCA}\}$ and maximizing human conditional Markovian syntax.

Construct	Codons	GC3	CpG/100	H_{cod} (b)	Cross d_{2b}	Cross d_{3b}	Cross S_{syn}	Int S_{syn} (S_{anti})	σ_{dup}	TTA/CTA/TCA | CAT	Sense stops (+0/+1/+2)	Antisense stops (−0/−1/−2)	Max anti span / Frame-−0 ATG→End
WT_SARS2_Spike_NC045512	1,274	0.2665	0.76	5.5173	+0.0222	+0.1029	+0.0619	+0.5134 (+0.4705)	17.8925	28/9/26 (63) | 13	1/55/145	63/64/134	106c (−1) / 79c (650..728)
BNT162b2_Full_Spike_2P	1,275	0.8105	5.52	4.9006	−0.0549	+0.2015	+0.0300	+0.2250 (+0.2165)	21.2378	0/0/0 (0) | **2**	2/59/20	**0**/43/72	**1,275**c (−0) / **1,101**c (175..1275)
mRNA1273_Full_Spike_2P	1,276	0.9552	7.21	4.4700	−0.0616	+0.3175	+0.1154	+0.1263 (+0.1168)	25.0785	1/1/2 (4) | **0**	3/76/**2**	4/36/54	458c (−0) / **0**c (0..0)
mRNA1273_NCBI_OR134578	1,273	0.9425	7.18	4.5336	−0.0635	+0.3064	+0.1059	+0.1468 (+0.1367)	24.3469	1/1/2 (4) | **0**	2/76/**5**	4/36/54	458c (−0) / **0**c (0..0)
Human_MaxCAI_Spike_2P	1,274	0.9670	6.60	4.1578	−0.0647	+0.3055	+0.1213	+0.0000 (+0.0000)	26.4177	0/0/0 (0) | 0	1/79/**0**	**0**/32/61	**1,274**c (−0) / 0c (0..0)
Human_MaxGC3_Spike_2P	1,274	1.0000	6.60	4.1578	−0.0666	+0.3248	+0.1339	+0.0000 (+0.0000)	26.4180	0/0/0 (0) | 0	1/79/**0**	**0**/32/61	**1,274**c (−0) / 0c (0..0)
Human_Markov_H2b_Spike_2P	1,274	0.6588	3.25	5.6133	**+0.2194**	+0.3250	+0.1553	+0.6342 (+0.5712)	19.7486	22/0/5 (27) | 9	1/41/40	27/54/86	197c (−0) / 103c (620..722)
Human_DuplexSafe_Viterbi_2P	1,274	0.6769	3.51	5.5910	**+0.2142**	**+0.3304**	**+0.1557**	+0.6190 (+0.5562)	19.7983	23/1/6 (30) | **0**	1/41/39	30/47/87	**174**c (−0) / **0**c (0..0)

Table 5: Six-frame duplex torus and Markovian syntax audit produced by `codon_torus_real_audit_v3_bnt162b2.py` (`v3.5`, zero simulations) across wild-type SARS-CoV-2 Spike (`NC_045512.2`), Pfizer `BNT162b2` (3,825 nt), Moderna `mRNA-1273` (3,828 nt and NCBI vector `OR134578.1`, 3,819 nt), deterministic human Max-CAI, deterministic human Max-GC3, unconstrained Viterbi Markov compilation, and the Corollary 4.4 Duplex-Safe Viterbi compiler scored under the `Human_T2T` reference kernel (2,342,893 held-out test transitions).

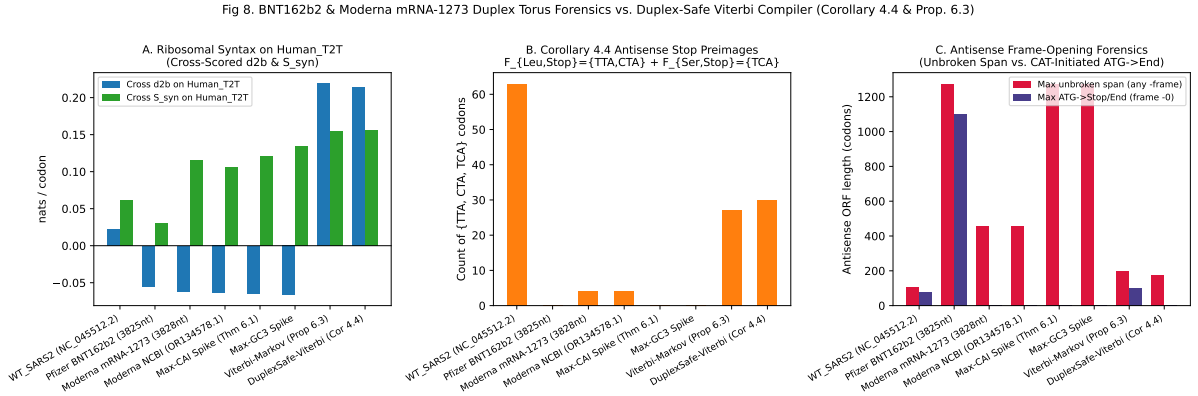


Figure 3: Simulation-free duplex torus and six-frame ORF forensics across the Spike-2P constructs (`codon_torus_real_audit_v3_bnt162b2.py`, `v3.5`): (A) cross-scored ribosomal syntax d_{2b} and S_{syn} on the `Human_T2T` reference kernel; (B) count of sense codons in the antisense termination preimage fibers $F_{\mathrm{Leu,Stop}} = \{\mathrm{TTA}, \mathrm{CTA}\}$ and $F_{\mathrm{Ser,Stop}} = \{\mathrm{TCA}\}$ (Corollary 4.4); (C) longest unbroken antisense open reading span (crimson) versus CAT-initiated frame-−0 ATG → Stop/End ORF length (navy) in codons.

Table 5 and Figure 3 establish four exact mathematical and bioengineering results:

1. **Antisense stop-layer annihilation and the** 1,101**-codon (3,303-nt)** CAT**-initiated antisense ORF in Pfizer** `BNT162b2`**:** In wild-type SARS-CoV-2 Spike, the three sense codons of $\rho^{-1}(\alpha^{-1}(\mathrm{Stop})) = \{\mathrm{TTA}, \mathrm{CTA}, \mathrm{TCA}\}$ occur 63 times (TTA = 28, CTA = 9, TCA = 26), placing 63 in-frame termination codons along antisense frame −0 (4.95 stops/100 codons) and restricting the longest unbroken antisense span to 106 codons (318 nt, frame −1; 103 codons / 309 nt in frame −0) and the longest frame-−0 ATG → Stop antisense ORF to 79 codons (237 nt, triplets 650 . . . 728). In `BNT162b2`, GC3-rich optimization (GC3 = 0.8105, CpG density rising 7.27× from 0.76 to 5.52 per 100 nt) replaces all Leucine and Serine codons in $F_{\mathrm{Leu,Stop}} \cup F_{\mathrm{Ser,Stop}}$ (TTA = CTA = TCA = 0). By Corollary 4.4(i), this deletes all 63 termination codons in antisense frame −0, converting the entire 1,275-codon (3,825 nt) antiparallel frame into a single unbroken open reading span. Crucially, because `BNT162b2` retains 2 sense CAT Histidine codons near its 3′ end, ρ(CAT) = ATG supplies an in-frame antisense initiation site at antiparallel triplet 175 that remains open for 1,101 consecutive codons (3,303 nt, triplets 175 . . . 1275).

2. **The complementary Corollary 4.4 regime in Moderna `mRNA-1273` (CAT = 0 antisense guard versus 820-codon sense frame-+2 opening):** Both the monovalent 3,828-nt Moderna `mRNA-1273` sequence (with triple terminal stop `TGATAATAG`) and the 3,819-nt NCBI expression vector `OR134578.1` exhibit even more extreme wobble-G/C enrichment (GC3 = 0.9552 and 0.9425; CpG density rising 9.50× to 7.21 and 7.18 per 100 nt) and replace every Histidine codon with CAC (#{CAT} = 0). Confirming Corollary 4.4(i),(iii), eliminating sense CAT drives the number of in-frame ATG start codons and ATG → Stop/End ORFs in antisense frame −0 identically to **zero** (0 codons vs. 1,101 in `BNT162b2`), while 4 retained {TTA, CTA, TCA} codons (1/1/2) partition antisense frame −0 into segments of at most 458 codons (1,374 nt, triplets 598 . . . 1055). However, because Moderna's GC3 = 0.9552 design depletes wobble-T sense codons from 590 (wild-type) and 139 (`BNT162b2`) down to just 33 across the entire gene, Corollary 4.4(ii) manifests on the sense strand: the number of termination codons in shifted **sense frame** +2 collapses 72.5-fold from 145 (wild-type, max unbroken span 42 codons) to just **2** in monovalent `mRNA-1273` (5 in `OR134578.1`), opening an unbroken stop-free span of **820** codons (2,460 nt, triplets 122 . . . 941) in sense frame +2.
3. **Negative Max-CAI/GC3 cross-scoring penalty and duplex strand-current divergence σ_{dup}:** Scored under the `Human_T2T` transition kernel P_{WT}, wild-type Spike exhibits positive conditional codon-to-amino-acid context ($d_{2b} = +0.0222$ nats/codon), whereas Pfizer `BNT162b2` ($d_{2b} = -0.0549$), Moderna `mRNA-1273` ($d_{2b} = -0.0616$), Moderna `OR134578.1` ($d_{2b} = -0.0635$), deterministic Max-CAI ($d_{2b} = -0.0647$), and deterministic Max-GC3 ($d_{2b} = -0.0666$) all incur the strictly negative Kullback–Leibler cross-scoring penalty proved in Proposition 6.2, while internal S_{syn}, S_{anti}, d_{2b}, and L^2 synonymous fiber variance $\mathbb{E}_P[\text{Var}_P(s(C) \mid A)]$ vanish identically (0.0000) on Max-CAI and Max-GC3 (Theorems 6.1 and 11.2). Simultaneously, the duplex strand-current divergence $\sigma_{\text{dup}} = D_{\text{KL}}(J_f \parallel J_\rho)$ (Proposition 8.5) escalates from 17.8925 nats in wild-type Spike to 21.2378 nats in `BNT162b2`, 25.0785 nats in `mRNA-1273`, and 26.4180 nats in Max-GC3.
4. **Constructive resolution via the Corollary 4.4 Duplex-Safe Viterbi compiler:** Our deterministic zero-RNG Viterbi compiler `Human_DuplexSafe_Viterbi_Spike_2P` demonstrates that positive human ribosomal syntax and six-frame duplex closure are mutually compatible. By enforcing $\phi(\text{His}) = \text{CAC}$ (#{CAT} = 0) alongside periodic {TTA, CTA, TCA} barrier retention within the dynamic-programming trellis of Proposition 6.3, it encodes the exact same Spike-2P protein while simultaneously achieving: (i) strongly positive human ribosomal syntax ($d_{2b} = +0.2142$, $S_{\text{syn}} = +0.1557$, $d_{3b} = +0.3304$ nats/codon) and near-native duplex divergence ($\sigma_{\text{dup}} = 19.7983$ nats); (ii) **zero** ATG → End ORFs in antisense frame −0 (CAT = 0, curing Pfizer's 1,101-codon antisense ORF); (iii) 30 in-frame termination barriers in antisense frame −0 (TTA = 23, CTA = 1, TCA = 6), capping the longest unbroken antisense span at 174 codons (522 nt); and (iv) 39 termination barriers in shifted sense frame +2, capping the longest frame-+2 span at 144 codons (432 nt, curing Moderna's 820-codon frame-+2 span).

13 Unified summary of proved theorems and explicit non-claims

We summarize the deductive chain established in this paper:

P1. Under a base encoding $B \cong \mathbb{Z}/4\mathbb{Z}$, the codon space is the discrete torus $\mathbb{T}_4^3 = (\mathbb{Z}/4\mathbb{Z})^3$, equipped with the mutational Hamming metric d_H and the auxiliary cyclic Lee metric d_{Lee} (Proposition 2.2).

P2. Watson–Crick reverse complementation $\rho(c) = -Rc + \mathbf{31} \pmod 4$ is a fixed-point-free affine involution isometric in both d_H and d_{Lee}, partitioning $\mathbb{T}_4^3$ into 32 mirror orbits and decomposing the ambient frequency space $\mathbb{R}^{64} = V_+ \oplus V_-$ into 32-dimensional mirror-even and mirror-odd subspaces (Theorem 2.5 and Corollary 2.6).

P3. On the abelian group $\mathbb{T}_4^3$, the 64 characters $\chi_k(c) = i^{k \cdot c}$ diagonalize translation-invariant kernels, and mirror-equivariant kernels block-diagonalize as $T_P = T_+ \oplus T_-$ on $V_+ \oplus V_-$

(Proposition 2.7).

P4. Coordinate closure modulo 4 on $\mathbb{T}_4^3$ does not identify start (ATG) and stop (TAA, TAG, TGA) codons (Proposition 3.3), whereas every canonical coding sequence satisfies an exact reading-frame phase closure modulo 3 ($t \equiv s \pmod 3$), Proposition 3.4), and closed translational cycles with $\sigma_{\text{cycle}} > 0$ arise when an explicit ribosome-recycling edge $\top_{\text{stop}} \to \bot_{\text{start}}$ is adjoined (Proposition 3.5).

P5. The single-strand genetic code fibration $\alpha : \mathbb{T}_4^3 \to \mathcal{A}$ ($64 \to 21$) defines a product of synonymous probability simplices Δ_{syn} of exact real dimension $64 - 21 = 43$ ($61 - 20 = 41$ on sense codons), whereas the two-strand duplex fibration $\alpha_{\text{dup}}(c) = (\alpha(c), \alpha(\rho(c)))$ partitions $\mathbb{T}_4^3$ into 48 singletons and 8 binary doublets (56 joint sense–antisense channels), reducing the duplex synonymous dimension to $64 - 56 = 8$ (7 on sense codons) and splitting $H(C \mid A) = H(A^\rho \mid A) + H(C \mid A, A^\rho)$ and identifying the 3-codon antisense termination preimage $\rho^{-1}(\alpha^{-1}(\text{Stop})) = \{\text{TTA}, \text{CTA}, \text{TCA}\}$ and initiation preimage $\rho^{-1}(\{\text{ATG}\}) = \{\text{CAT}\}$ (Proposition 4.1, Remark 4.2, Theorem 4.3, and Corollary 4.4).

P6. Without any stationarity assumption, total codon-bigram mutual information decomposes into additive non-negative chain-rule layers via the tripartite identity (2) and the quadripartite identity (4), isolating $d_{2b} = I(C_{i-1}; A_i \mid A_{i-1})$, $S_{\text{syn}} = I(C_{i-1}; C_i \mid A_i) = d_2 - d_{2b}$, and $S_{\text{syn}}^{(2)} = I(C_{i-1}; C_i \mid A_{i-1}, A_i) = d_{2,\text{pair}} - d_{2b}$ while total bigram information is invariant under antiparallel duplex inversion, $I(\rho(C_i); \rho(C_{i-1})) = I(C_{i-1}; C_i)$, and S_{syn} splits into the antisense amino-acid program $S_{\text{anti}} = I(C_{i-1}; A_i^\rho \mid A_i)$ (74.1%–88.5% across nine genomes) plus the 8-doublet residual $S_{\text{dup}} = I(C_{i-1}; C_i \mid A_i, A_i^\rho)$ (Theorems 5.2, 5.3, 5.4, and 5.5).

P7. The projected amino-acid process $A_i = \alpha(C_i)$ is a Kemeny–Snell lumped first-order Markov chain of the codon process if and only if $I(C_{i-1}; A_i \mid A_{i-1}) = d_{2b} = 0$ (Theorem 5.6).

P8. Any deterministic Max-CAI-type design $C_i = \phi(A_i)$ collapses $I(C_{i-1}; A_i \mid A_{i-1}) = I(A_{i-1}; C_i \mid A_i) = S_{\text{syn}} = S_{\text{syn}}^{(2)} = 0$, is invariant under intra-gene synonymous multiset shuffles, and incurs a non-positive KL cross-scoring penalty $d_{2b}(\phi; P_{\text{WT}}) \le 0$ under a natural reference kernel, whereas the conditional synonymous kernel $P_{\text{WT}}(C_i \mid C_{i-1}, A_i)$ is variationally optimal (Theorem 6.1, Propositions 6.2–6.3, and Section 12.1).

P9. Downstream physicochemical predictability obeys the data-processing bound $I(C_{i-1}; Q_i \mid A_{i-1}) \le I(C_{i-1}; A_i \mid A_{i-1}) = d_{2b}$ (Theorem 7.1).

P10. Any stationary Markov chain on $\mathbb{T}_4^3$ satisfies $\sigma = D_{\text{KL}}(J_f \parallel J_r) \ge \frac{1}{2}\|J_f - J_r\|_1^2 \ge 0$, with $\sigma = 0$ if and only if detailed balance (zero cycle affinity on all loops) holds, $\sigma_{\text{exact}} < +\infty$ if and only if $F_{\text{unidir}} = 0$, and dynamic mirror balance implies static Chargaff mirror symmetry $\pi \in V_+$ with duplex strand-asymmetry divergence $\sigma_{\text{dup}} = D_{\text{KL}}(J_f \parallel J_\rho) \ge \frac{1}{2}\|J^{\text{mir}}\|_1^2 \ge 0$ vanishing if and only if $J^{\text{mir}} = 0$ (Theorem 8.1 and Propositions 8.2–8.5).

P11. In $L^2(\mathcal{C}, P)$, the conditional expectation operator Π_α is the orthogonal projector onto the 21-dimensional amino-acid pullback space $\text{Im}(\alpha^*)$ with 43-dimensional kernel, while window marginal aggregation erases S_{syn} (Theorem 11.2 and Lemma 11.3).

P12. The conditional independence null hypotheses $S_{\text{syn}} = 0$, $S_{\text{syn}}^{(2)} = 0$, and $d_{2b} = 0$ admit simulation-free analytic G^2, $G_{(2)}^2$, and G_{lump}^2 tests with exact degrees of freedom df = 2709 ($64 \to 21$), 2460 ($61 \to 20$), 1849 (43^2), 1681 (41^2), 860 (43×20), and 779 (41×19), corroborated across 9,834,024 held-out RefSeq transitions in nine reference genomes (Proposition 9.1, Remarks 9.2–9.3, and Section 12).

Conversely, to prevent any numerological misinterpretation:

N1. We do not claim that the physical DNA double helix is a spatial torus in $\mathbb{R}^3$.

N2. We do not attribute any cosmological, particle-physics, or mystical meaning to the integers 2, 3, 4, 6, 7, 8, 16, 19, 20, 21, 24, 32, 40, 41, 42, 43, 48, 54, 56, 60, 61, 62, 63, 64, 361, 400, 779, 860, 1681, 1849, 2460, or 2709.

N3. We do not use Heegner numbers, kissing numbers, algebraic number fields ($\sqrt{2}, \sqrt{3}, \sqrt{5}$), periodic-table closures, or geometric hourglasses in any definition or proof.

N4. We do not claim that start and stop codons are identical triplets on the torus; only their

reading-frame phase modulo 3 is conserved.

N5. We do not rely on Monte Carlo shuffling as a mathematical proof; all theorems are proved deductively and paired with analytic contingency and held-out likelihood audits on real biological databases.

14 Conclusion

By taking the antiparallel DNA duplex $\mathcal{D} = \{(c, \rho(c)) : c \in \mathbb{T}_4^3\}$ on the finite discrete torus $\mathbb{T}_4^3 = (\mathbb{Z}/4\mathbb{Z})^3$ as the primitive state space of protein-coding sequences, this paper unifies static genomic symmetry and dynamic sequential syntax within a single parameter-free algebraic and information-theoretic framework:

1. **Duplex geometry and the form–program split:** The Watson–Crick reverse-complement operation $\rho(c) = -Rc + \mathbf{31} \pmod 4$ is a fixed-point-free affine involution isometric in both the Hamming and Lee metrics (Theorem 2.5), partitioning the 64 codons into 32 mirror orbits and decomposing the frequency space orthogonally as $\mathbb{R}^{64} = V_+ \oplus V_-$ ($\dim_{\mathbb{R}} V_+ = \dim_{\mathbb{R}} V_- = 32$; Corollary 2.6). The mirror-even subspace V_+ governs static Chargaff duplex parity (*form*) [3, 4, 5, 16, 17], whereas the mirror-odd subspace V_- and non-equilibrium Markov transition currents govern strand-asymmetric coding bias (*program*) [23].
2. **Two-strand genetic code fibration and the 8-doublet theorem:** Whereas the single-strand genetic code surjection $\alpha : \mathbb{T}_4^3 \to \mathcal{A}$ ($64 \to 21$) leaves a 43-dimensional synonymous simplex bundle (41 on sense codons; Proposition 4.1), evaluating the genetic code on both strands via $\alpha_{\mathrm{dup}}(c) = (\alpha(c), \alpha(\rho(c)))$ maps the wobble base of the sense codon onto the leading base of the antisense codon, partitioning $\mathbb{T}_4^3$ into 48 singletons and 8 binary doublets (56 joint channels) and reducing the duplex synonymous fiber dimension from 43 to $64 - 56 = 8$ (7 on sense codons; Theorem 4.3).
3. **Antisense channel dominance of the synonymous program:** Across all nine reference genomes from NCBI RefSeq (9,834,024 held-out codon transitions spanning Eukarya, Bacteria, and Archaea), the exact two-strand decomposition $S_{\mathrm{syn}} = S_{\mathrm{anti}} + S_{\mathrm{dup}}$ (Theorem 5.5) reveals that 74.1%–88.5% of the sense-strand contextual synonymous program $S_{\mathrm{syn}} = I(C_{i-1}; C_i \mid A_i)$ is carried directly by the antisense amino-acid channel $A_i^\rho = \alpha(\rho(C_i))$ ($S_{\mathrm{anti}} = I(C_{i-1}; A_i^\rho \mid A_i)$; Table 4).
4. **Markov lumpability, Max-CAI collapse, and non-equilibrium dynamics:** In all nine taxa, Kemeny–Snell lumpability of the amino-acid projection is rejected ($d_{2b} = I(C_{i-1}; A_i \mid A_{i-1}) > 0$, df $= 779$, $p < 10^{-300}$; Theorem 5.6), downstream physicochemical predictability obeys the data-processing inequality $0 < I(C_{i-1}; Q_i \mid A_{i-1}) < d_{2b}$ across all 36 tests (Theorem 7.1), deterministic Max-CAI recoding ($C_i = \phi(A_i)$) annihilates $S_{\mathrm{syn}}, S_{\mathrm{syn}}^{(2)}, d_{2b}$ to zero while incurring a negative Kullback–Leibler cross-scoring penalty in 77.0%–98.7% of genes (Theorem 6.1 and Proposition 6.2), and codon-torus Markov chains exhibit strictly positive entropy production ($\sigma_{\mathrm{reg}} > 0$) and directed duplex mirror currents ($\|J^{\mathrm{mir}}\|_1 > 0$; Theorem 8.1 and Propositions 8.2–8.5).

Together with the L^2-orthogonal projection theorem (Theorem 11.2) and Lemma 11.3, these results explain why macroscopic genomics–proteomics hourglass concordance on chromosomal windows [18, 22] coexists with a rich, non-equilibrium first-order synonymous program on the double-stranded codon torus.

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